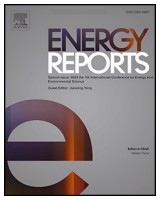
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Review article



Intelligent industrial demand response to increase grid flexibility and reliability: A review

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ABSTRACT

The rapid transition toward renewable energy has introduced challenges in grid stability due to the intermittency of non-dispatchable sources like solar and wind. Industrial Demand Response (IDR) offers a promising, cost-effective solution that adjusts energy consumption patterns to align with supply, increases renewable utilization, and reduces costs. This review provides an updated analysis of IDR, sorting technologies into five categories: energy storage, scheduled energy usage, operational flexibility, on-site generation, and intelligent operations. Energy storage solutions, while requiring little flexibility, often have the longest payback periods. While slightly better, on-site generation also has longer payback periods, ranging from 5 to 20 years or more. Scheduled energy usage, operational flexibility, and intelligent operations allow significant peak reduction at lower capital costs but require greater flexibility. While 15–20 % peak reduction is within the range of all five categories, scheduled energy use and on-site energy generation are shown to have reductions of up to 70–80 % in select scenarios. Combining multiple IDR strategies from these five categories maximizes both financial and operational benefits. Synergistic approaches are shown to enhance grid stability while reducing costs. As the grid evolves, IDR will enable a more flexible, renewable-powered future that will benefit industrial facilities and the broader energy system.

1. Introduction

The transition to cleaner energy generation has driven a rapid increase in renewable energy penetration. Over the past decade, the electricity grid has undergone significant transformation, with projections indicating continued growth over the next 30 years (What the duck curve tells us about managing a green grid, 2016). However, this progress brings challenges in maintaining grid stability and reliability. Many renewable energy sources, such as solar and wind, are considered to be non-dispatchable. Inherently intermittent and weather-dependent, these energy sources have more unpredictable power output. Excessive variable generation is currently curtailed during off-peak periods when there is not enough demand for electricity to match the supply (Paterakis et al., 2017). One example of this supply-demand mismatch between supply and demand becomes apparent when viewing the iconic "duck curve," first introduced in California in 2013. A high level of non-dispatchable solar energy has caused deep valleys and prominent peaks left to be generated by electricity providers. Demand-side flexibility is a promising, cost-effective alternative where energy consumption patterns are adjusted to align with availability and grid needs (Paterakis et al., 2017).

Demand Response (DR) has emerged as a critical tool for enabling this flexibility. DR is defined as "*changes in electric usage by end-use customers from their normal consumption patterns in response to changes in the price of electricity over time, or to incentive payments designed to induce lower electricity use at times of high wholesale market prices or when system reliability is jeopardized*" (Jordehi, 2019). This 'electric usage' refers to energy used beyond the meter, as this is the demand of the facility on the electricity grid. By incentivizing consumers to modify their electricity usage in response to grid signals, DR helps shift consumption away from peak periods and reduces strain on the grid (Leherbauer et al., 2025). Some primary tactics are shown in Fig. 1. Three of these solutions, valley filling, strategic conservation, and strategic load growth, are typically best addressed with energy management programs: programs that oversee a facility's energy use in the long term. The methods in this paper focus more on the other three solutions: peak clipping, load shifting, and flexible load shape. These solutions can often be addressed in part with one-time additions or changes. Responses to prices in electricity typically drive DR; energy is cheaper when it is

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Abbreviations

BESS	Battery Energy Storage System
CAES	Compressed Air Energy System
CHP	Combined Heat and Power
DEC	Dynamic Equipment Control
DR	Demand Response
DRP	Demand Response Program
IDR	Industrial Demand Response
MPC	Model Predictive Control
PAF	Predictive Analytics and Forecasting
PV	Photovoltaic
RTM	Real-Time Monitoring
RTS	Real-Time Scheduling
TES	Thermal Energy Storage
TOU	Time-Of-Use
VFD	Variable Frequency Drive

Definitions

Industrial demand response Strategically planning and/or changing an industrial process's electrical profile to keep peak electrical usage lower than regular operation.

Demand The time-dependent amount of power a facility uses.

Peak demand use The time of the day, week, or year when a facility uses the largest demand.

Flexible demand The ability to change when electricity is used throughout a day, week, or year.

Energy savings Savings obtained from a decrease in energy use or price.

Demand savings Savings obtained from a decrease in demand use or price.

Demand response participation A savings mechanism where a facility lowers its demand in response to a request from the electricity provider.

Time-of-use rates Electricity prices that change depending on the time of day, week, or year. This can happen as part of a prescribed plan or in real-time.

available and more expensive when electricity is in high demand (Wang et al., 2024).

While the industrial sector has been slower in receiving and implementing recommendations, manufacturers' participation is vital to unlocking the full potential of demand-side flexibility (Shoreh et al., 2016). In 2023, the EIA reported that manufacturers used 27 % of the U.S.'s electricity generation (U.S. energy consumption by source and sector, 2023). A more grid-responsive industrial sector could better utilize renewable energy, advance efforts to electrify energy generation, enhance grid stability, and reduce overall costs by creating a more flexible industrial demand (Nandy et al., 2022; Billings and Powell, 2022). The use of DR in industrial settings is called industrial demand response, or IDR.

Many reviews have analyzed the field by industrial facility type or sector (Paterakis et al., 2017; Shoreh et al., 2016; Jiang et al., 2019; Siddiquee et al., 2021; Ranaboldo et al., 2024). These reviews have focused on facility types or individual technologies, but none have systematically compared IDR strategies across categories with respect to payback, peak reduction, and synergy. This strategy of focusing on specific manufacturing areas aids the impact of IDR in larger or select industrial facilities, but it can seem exclusive to other facilities that do not have as much background or may be smaller by nature. To address these concerns, the main points of novelty of this review include:

- This study systematically categorizes IDR solutions based on their payback periods and impact on energy timing, highlighting the trade-offs between high-payback but low-impact solutions (e.g., energy storage) and low-payback but high-impact solutions (e.g., scheduling and operational flexibility).
- The research provides summaries of quantitative data obtained by IDR strategies, presented in tabular and graphical formats. While figures throughout the body of each section are pulled from notable papers used in the review, figures summarizing the data at the end of each section were developed by the authors. When a large quantity of data is available, the median and quartiles are presented to give a brief look at each technology. A more in-depth tabulation of data available and papers used is included in the supplementary material. This summarized data includes, when available:
  - o Percentages of peak demand reduced or shifted.
  - o Cost savings resulting from the change in peak energy usage.
  - o Implementation costs.
  - o Payback periods.
  - o Sizing and efficiencies where applicable.

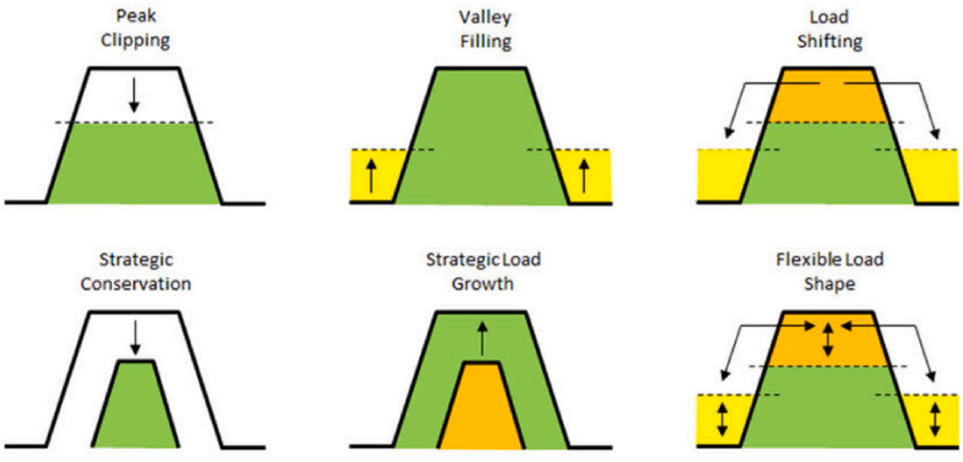


Fig. 1. Typical methods to address high peak demand, collectively referred to as demand response (DR). Image acquired from Domínguez-Garabitos et al (Domínguez-Garabitos et al., 2022).

- By numbering the frequency of combined technology implementations, this study emphasizes how integrating multiple IDR solutions leads to greater overall energy and cost savings, including specific combinations such as intelligent operations with scheduling or energy storage with generation.
- The study reinforces the role of IDR in supporting grid stability, renewable energy integration, and the electrification of energy generation, positioning IDR as a critical component of the future energy landscape.

The review proceeds as follows:

- Section 2 describes the methods used to collect papers and data on IDR as well as the standard method used to calculate and report payback periods.
- Section 3 covers storage technologies. These pieces of equipment store energy in various ways and discharge it when peak demand is high, lowering the electricity need from the utility provider.
- Section 4 details scheduling techniques. These techniques adjust operation in different ways to lower energy usage known peak demand times.
- Section 5 outlines technologies capable of implementing operational flexibility. These tactics take advantage of real-time pricing structures to lower energy usage in the moment it is needed.
- Section 6 describes a selection of onsite energy generation sources. While there are many ways a facility may generate its own electricity to lower the dependence on the grid, these five techniques are most common for IDR.
- Lastly, Section 7 summarizes intelligent operations. These ideas take advantage of software, models, and optimization techniques to further increase the impact of the aforementioned technologies.

2. Methods

2.1. Literature search

The literature search was conducted using Google Scholar, Scopus, and Academic Search Ultimate. Initial queries employed the keywords “Industrial Demand Response”, “Demand Response in Manufacturing”, “Industrial Demand Shifting”, and “Industrial Peak Shaving”. After the main structure of the review was established, these search terms were combined with the five categories highlighted in this paper—energy storage, scheduling, operational flexibility, on-site generation, and intelligent operations—to capture additional relevant publications that may not have appeared in the initial searches.

The search emphasized publications from 2014 to 2025 in order to reflect the most current developments in industrial demand response. Older studies were included only when they were exceptionally influential or demonstrated unique relevance to later work. All of the selected sources were drawn from peer-reviewed journals to maintain a consistent standard of quality and reliability.

Studies were considered eligible if they met three criteria. First, they had to focus on industrial facilities or rely on manufacturing load data rather than residential or commercial contexts. Second, they needed to report quantitative results on demand reduction or shifting, typically in units of kilowatts or kilowatt-hours. Third, the methods used to achieve these changes had to be described in sufficient detail to allow for a clear understanding of the underlying approach. While the inclusion of data on payback periods, monetary savings, equipment size, or facility type was not required, these details were extracted whenever available to enrich the analysis.

2.2. Economic results

The approach of using IDR at manufacturing sites treats industries as their own agents, charged for their electricity at the meter and typically unable to sell electricity back to the grid. In this way, IDR mechanisms are able to take advantage of two main sources of incentives to offer financial benefits. The first incentive is demand charges, charges based on the kW usage of a facility, such as maximum monthly demand, coincidental peak demand, or some demand response participation programs. The second is energy charges, charges based on the kWh usage of a facility, such as other demand response participation programs, real-time pricing, or time-of-use rates (Nandy et al., 2022; Nezamoddini and Wang, 2017). While some scenarios would have other economic benefits, such as avoiding black-outs, brown-outs, or other results of grid instabilities, there are more difficult to quantify and are not typically included in economic analysis of IDR. The economic benefits of IDR were calculated according to the following hierarchy:

- The savings for these studies were calculated by summing up the demand and energy charges, implementing the savings mechanism, subtracting the new demand and energy charges, and adding any incentives received. If the costs for the study were reported, the simple payback period was calculated by dividing the project costs by the yearly savings.
- If there was not enough data for this analysis, the payback period stated in the research paper was used. This method came second because of the variety of ways to calculate a payback period over the diverse background of papers. However, this method was needed due to the absence of needed data to calculate the payback period independently.

Because the results reviewed in this work are from a diverse background, normalization of numbers was needed. The standard monetary choice is the 2024 U.S. dollar. If original results were from before 2024 or in another currency, they have been updated using conversion and inflation rates to reflect the 2024 U.S. dollar. This was done according to data from the U.S. Bureau of Labor Statistics and the U.S. Treasure Fiscal Data. Key reported metrics include quantity and percentage of demand and energy shifted or reduced. The quantity is a measurement of the lowered amount of demand or energy during an IDR time, whether that was during the facility’s peak time, the utility’s peak time, a time of high energy pricing, or during DRP events. The percent reduction is the amount of demand or energy shifted divided by the original demand or energy use during the above events.

The context of the data included in this review can be at times highly dependent on load profile, industry type, location, and pricing structure. However, this level of detail conflicts with the intent to give a general overview of each technology and obtainable results possible with each solution. More detailed information, including the industry type, pricing structure, and geographical location, is included in the Supplementary Material.

3. Storage: active technologies to shift demand

Energy storage technologies are a popular IDR solution, storing energy during off-peak hours and discharging it during on-peak hours, thereby allowing a decoupling of grid supply and process demand. This energy can be stored in many forms: chemical, electrochemical, thermal, compressed air, potential, and kinetic. This storage allows peak reduction to occur with little to no disturbance or changes to the process. However, these technologies typically come with higher capital costs, requiring high incentives or savings to make them economically feasible. Using storage in IDR situations can have a very different effect when compared to commercial or residential scenarios. That is because industrial load profiles often have peaks at different times of the day and have larger peak demands, resulting in unique and meaningful savings mechanisms.

3.1. Electrochemical batteries

Electrochemical batteries, particularly lithium-ion types, are a key technology enabling industrial processes to shift energy usage and reduce demand during peak hours, thus optimizing energy costs. Through high round-trip efficiency, these batteries offer significant savings opportunities across various strategies, including peak shaving, time-of-use (TOU) rates, and participation in demand response programs. As demonstrated in Fig. 2, the integration of battery energy storage systems (BESS) in peak shaving strategies smooths out fluctuations in energy demand, with the BESS charging during off-peak hours and discharging during peak hours, reducing peak load but increasing total energy usage because of storage inefficiencies.

Lithium-ion batteries have emerged as a preferred technology for grid-scale energy storage due to their high round-trip efficiency (95 %) (Chen et al., 2020; Peng et al., 2020). Depending on facility needs, these BESSs can be used with various discharge rates, ranging from hundreds of kilowatts to single-digit megawatt systems (Peng et al., 2020; Billings et al., 2022a, 2022b; Wang and He, 2016). However, the savings for BESS are not always enough to pay for the capital investment. This is due to high prices of BESSs, low economic incentives for peak demand reduction, or both. While a 6.7-year payback is reported in one case (Dougherty et al., 2021), other situations often report payback years of 10, 20, or well past the lifespan of a battery (Dougherty et al., 2021; Elio and Milcarek, 2023; Elio et al., 2022; Kozadajevs et al., 2018). These BESSs can be supplemented by model predictive control (MPC) and predictive analytics and forecasting (PAF), resulting in peak shavings of 20 % or 7–9-year payback periods (Zhang et al., 2016a; Wohlgenannt et al., 2024). These scenarios outline the typical 3–15 % peak reduction capable of BESSs.

It is very popular to accompany BESS with photovoltaic (PV) cells. Typically, PV arrays are oversized to sell energy back to the grid. BESS eliminates this need to sell electricity to the grid, which will be repurchased later at higher rates. Because savings can be obtained through reduced load and energy usage, the payback is reduced to 13–17 years (Paul et al., 2019; Chen et al., 2022). One advantage of this method is the ability to completely displace demand during demand response requests (Weitzel and Glock, 2019). These combined BESS and PV systems produced peak shavings of anywhere between 5 and 1660 kW, 5–60 % of the peak load (Chen et al., 2022; Fleschutz et al., 2022; Nan et al., 2021; Kusakana, 2020).

Table 1
Peak Reductions, Payback Periods, Size, and Synergistic Technologies of BESS from Select Studies.[1]

Papers used	(Peng et al., 2020; Billings et al., 2022a; Wang and He, 2016; Billings et al., 2022b; Dougherty et al., 2021; Elio and Milcarek, 2023; Elio et al., 2022; Kozadajevs et al., 2018; Zhang et al., 2016a; Wohlgenannt et al., 2024; Paul et al., 2019; Chen et al., 2022; Weitzel and Glock, 2019; Fleschutz et al., 2022; Nan et al., 2021; Kusakana, 2020; Yun et al., 2022; Ruiz Duarte et al., 2020; Musabandesu and Loge, 2021)	
	Median	IQR
Demand Shifted or Reduced (kW)	210	82.5 – 1,110
Percent of Demand Shifted or Reduced (%)	8.2	4.8 – 30
Energy Shifted or Reduced (kWh)	550	117.5 – 2,400
Percent of Energy Shifted or Reduced (%)	3.8	1.9 – 5.7
Payback Period (years)	13.8	9.5 – 19.5
Monetary Savings (USD per year)	16,000	4,413 – 37,320
Percent of Monetary Savings (%)	4.65	2.7 – 12
Battery Size (kWh)	1,000	353 – 7,250
Max Output (kW)	500	142.5 – 2,450

[1] This and subsequent result tables are the summary of a more detailed analysis available in the Supplementary Material.

BESS offers a versatile solution for IDR, with applications ranging from large-scale manufacturing to smaller operations. Table 1, a summary of the tabulated data for BESS, outlines the ranges of peak reduction, payback periods, and sizes of BESS currently in literature. BESS applications are helpful in a large range of demand sizes. However, the percentage of peak reduction is rarely above 20 %. Additionally, it shows the lengthy payback periods reported in most scenarios. Despite its high cost, BESS remains a viable option for any facility seeking to reduce peak load and improve grid stability, given its adaptability to different operational settings.

Because the application of BESS or BESS and PV is common in IDR, more data is available for this section and presented in Fig. 3 and Fig. 4. These figures compare the percentage of peak demand shifted to the battery discharge rate and size, respectively. They also incorporate

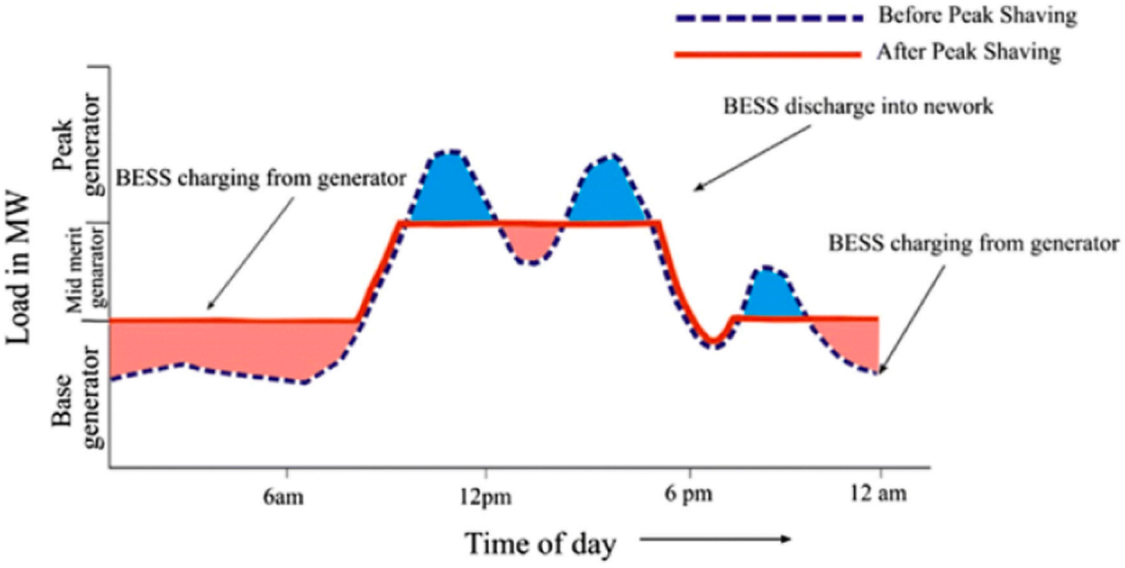


Fig. 2. Depiction of the typical use of BESS to store energy during off-peak hours to be discharged during on-peak hours to lower peak usage. Image acquired from Uddin et al (Uddin et al., 2018).

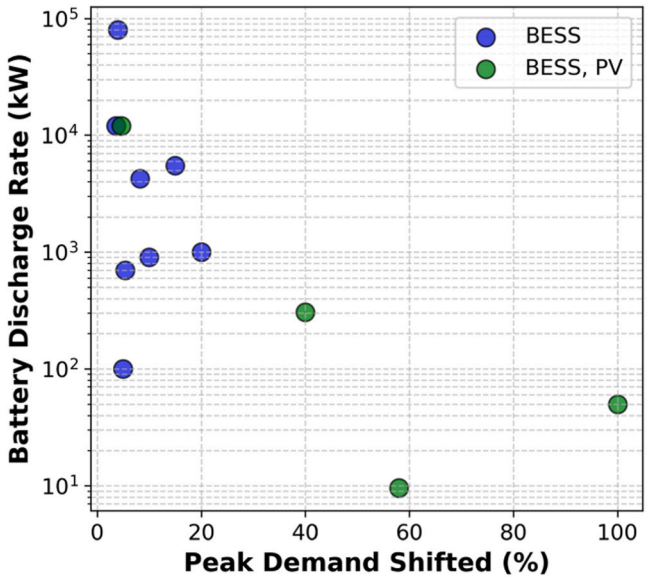


Fig. 3. Battery discharge rate vs. the percentage of the peak demand shifted for the compiled papers. Original image.

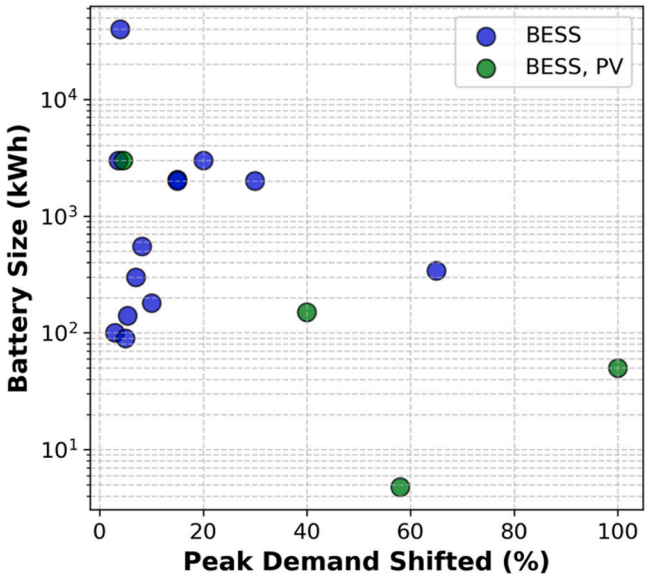


Fig. 4. Battery size vs. the percentage of the peak demand shifted for the compiled papers. Original image.

whether or not the battery was used in addition to PV cells. Note that the y-axes are logarithmic. The figures show that smaller batteries achieve larger percentages of demand shifting. Because the size of the battery limits the maximum demand saving, batteries are very effective at managing demand when the original demand is small. In larger facilities, the percentage of peak shavings is much smaller. In all scenarios, it is difficult to achieve a shift of over 20 % without PV cells.

3.2. Thermal energy storage

Thermal energy storage (TES) supports IDR by storing thermal energy to shift electrical loads. They store excess thermal energy during off-peak hours for use during high-demand periods, whether as hot or cold storage. Fig. 5 shows an example of how solar thermal storage may be integrated into an industrial process requiring hot water, though this heating or cooling could also be electrically provided. While heating loads are still met mainly with fossil fuels (45 % percent of industrial energy loads are from coal and natural gas (U.S. energy consumption by source and sector, 2023)), TES will become more important as electrification grows. Additionally, TES can be widely implemented because most industrial processes have heating and/or cooling loads.

The use of TES can result in demand reductions ranging from 0.2 to 4 MW, accounting for anywhere from 10 % to 50 % of the peak demand (Wohlgenannt et al., 2024; Cirocco et al., 2022; Molavi and Ardehali, 2016; Machalek and Powell, 2019a; Huang et al., 2017). Many facilities already have some volume for heated or cooled fluid storage. Utilizing this volume for thermal storage makes TES a cost-effective option. Its capital costs are typically lower than BESS, resulting in either a shorter payback period when minimal additional equipment is required or a 6–8-year payback when new infrastructure is needed (Machalek and Powell, 2019a; Bahr et al., 2023).

Similar to the reasoning behind BESS and PV, integrating TES with solar PV systems optimizes peak electricity demand management as well. These systems have been shown to reduce demand use by 20 %, energy usage by 10–30 %, and costs by 10 %. However, when combined with PV, the payback significantly increases to 20 + years, resulting in the same problems encountered with BESS + PV systems (Saffari et al., 2018; Arteconi et al., 2017). Because of its easy integration with facility heating and cooling loads, TES can become a part of large systems that use wind, solar, pumped water, and/or combined heat and power (CHP) solutions. These systems can see demand decrease on the order of megawatts, decreasing or exceeding the total peak demand by 50–150 % (Almehizia et al., 2019; Pei et al., 2019).

A summary of the tabulated data is located in Table 2, pulled from studies that best show key characteristics. Payback period is not included as the sources for price and savings were not the same; therefore, it would not be an accurate evaluation to calculate the payback manually. The results of TES are similar to those of BESS in that they are both very scalable, allowing for integration in many different facilities, but they are not often able to shift more than 20 % of peak demand. However, the financial commitment is not as significant as BESS, as shown through the above paybacks or the pricing in Table 2. TES offers versatile and practical solutions for IDR, providing cost savings while moving energy generation and storage toward electrification. Because of the widespread need for heating and cooling loads, TES is poised to play a critical role in many industrial facilities.

3.3. Compressed air storage

Compressed air energy storage (CAES) systems are a versatile and efficient option for IDR, offering advantages such as low operational costs, scalability, and high storage capacities. Unlike electrochemical batteries, CAES often avoids additional energy conversion steps, making it a cost-effective solution for load shifting and aligning energy demand with renewable energy availability (Boehm and Franke, 2017). Often, utility-scale CAES uses underground salt caverns to store large amounts of compressed air, as shown in Fig. 6. While industrial-sized CAES may use a similar configuration, the storage mechanism is typically a large tank rather than an underground cavern. CAES can be effective as compressed air is used in most facilities and accounts for roughly 10 % of electricity used in manufacturing in the U.S (Collins et al., 2001).

Considering CAES alone, real-time control of compressed air generation integrated with manufacturing processes can enhance energy flexibility without compromising throughput. Beier et al. show that the system shifts loads effectively to align with renewable energy availability, achieving a 9.6 % energy reduction while maintaining consistent production. This approach offers an efficient alternative to batteries and supports scalability for broader industrial use cases (Beier et al., 2015). Another study compares CAES with lithium-ion batteries, highlighting CAES's ROI (though still long, at 13.8 years). In one case, a CAES system with 333.3 kW power and 5000 kWh capacity achieved a similar peak reduction of 18 %, performing similarly to lithium-ion batteries in

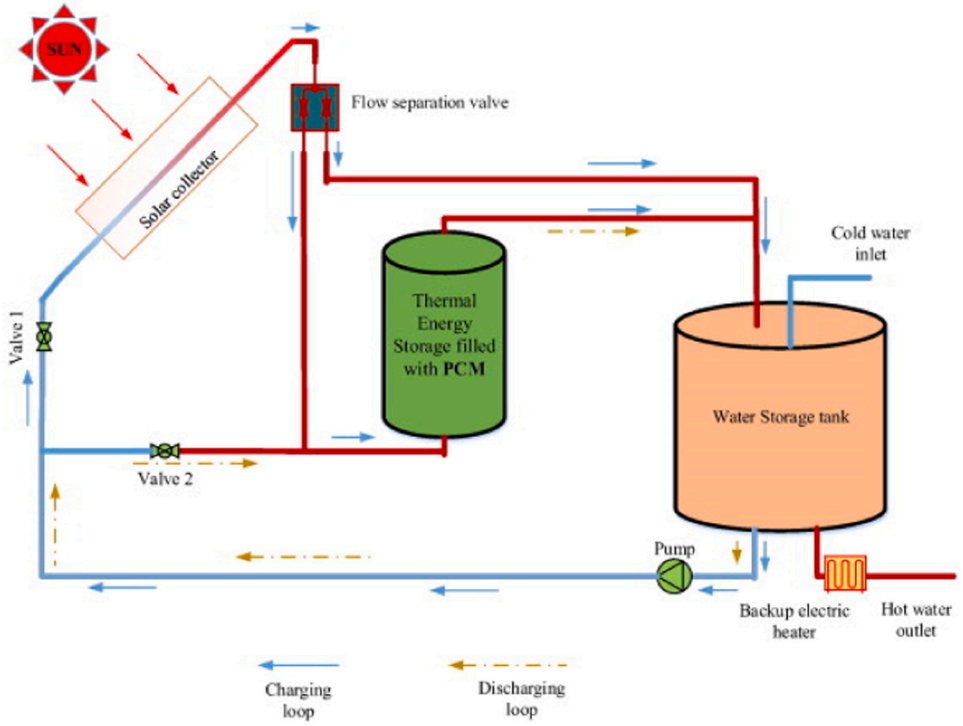


Fig. 5. One use of TES for hot water demand using a phase change material. While the heat source is the sun in this situation, many other sources can provide heat (or cooling) for TES. Image acquired from Mahfuz et al (Mahfuz et al., 2014).

Table 2
Peak Reductions, Pricing, Size, and Synergistic Technologies of TES from Select Studies.

Papers used	(Wohlgenannt et al., 2024; Cirocco et al., 2022; Molavi and Ardehali, 2016; Machalek and Powell, 2019a; Huang et al., 2017; Bahr et al., 2023; Saffari et al., 2018; Arteconi et al., 2017; Almehizia et al., 2019; Pei et al., 2019)	
	Median	IQR
Demand Shifted or Reduced (kW)	100	50 – 200
Percent of Demand Shifted or Reduced (%)	11	8 – 22
Energy Shifted or Reduced (kWh)	220	175–400
Percent of Energy Shifted or Reduced (%)	8	0.5 – 17
Monetary Savings (USD per year)	15,150	4,890 – 26,250
Percent of Monetary Savings (%)	8.55	3.15 – 11
Battery Size (kWh)	650	300 – 800
Price (USD)	161,000	7,030 – 41,400

cost-effectiveness and capacity (Elio et al., 2022).

CAES can also be combined with other process or technology improvements to further increase peak shaving, whether those systems are small or large. In smaller, 4–6 kW applications, Beier et al. and Diekerhof et al. studied CAES effectiveness when combined with scheduled manufacturing, effectively lowering peak demand (Beier et al., 2015; Diekerhof et al., 2016). In larger applications, CAES can become a part of systems using multiple technologies. Pei et al. studied CAES integration with thermal energy storage, CHP equipment, and predictive analytics in a lithium battery manufacturing plant, achieving 156 % energy reduction relative to baseline demand (Pei et al., 2019).

Table 3 summarizes the data from related papers. Because there are fewer sources for this strategy, ranges are reported rather than median and quartiles. CAES shows the continued trend that storage systems have difficulty lowering peak demand by more than 20 %. While slightly cheaper than BESS, the payback period is still long and may outlast the lifetime of the equipment.

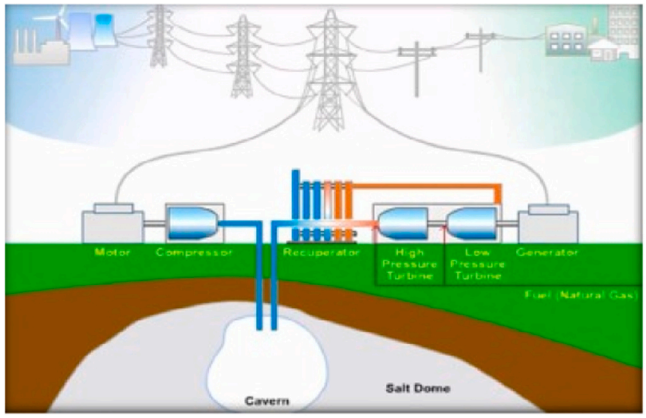


Fig. 6. A typical utility-sized CAES system, capable of storing excess compressed air during times of high demand. Image acquired from Benato and Stoppato (Benato and Stoppato, 2018).

3.4. Pumped water storage

Pumps and their required motors are another large energy user in facilities. In the U.S. industrial sector, pumping systems account for 25 % of the energy consumed by electric motors and over 50 % of the electricity in pumping-intensive industrial facilities (Pump Systems Matter, 2007). There are many publications on the possibility of storing and using potential energy by pumping water to a higher elevation to generate energy through a turbine at a desired time (Naval et al., 2023). One such example in industrial application is provided by Fekri et al., where a power plant tasked with providing power for industrial responsive loads can lower peak production by 70 MW, 7.8 % of the total demand (Fekri Moghadam et al., 2015). However, this is not as

Table 3
Typical Size, Payback Period, and Synergistic Technologies of CAES.

Paper Title Largest Peak Reduction	Demand Shifting Method	kW Shifted /Reduced	kWh Shifted /Reduced	Savings Mechanism	Payback Period (years)	Monetary Savings (per year)	Size
(Beier et al., 2015) (Elio et al., 2022)	Compressed Air Storage	**9.6, 18 %**	1,500	Demand costs, energy costs			333.3 kW / 5,000 kWh
Payback Period (Elio et al., 2022)	Compressed Air Storage	6.4, 18 %	150, 1,500	Demand costs, energy costs	**13.8, 18.6**	$39,000–77,000	166.7 kW / 2,500 kWh
CAES + Scheduling (Beier et al., 2015) (Diekerhof et al., 2016)	Compressed Air Storage, Scheduling Production /Curtailment	6.5, 22 %	2.3, 49 %	Energy costs		$10,500	4–10 kW

feasible for typical industrial processes at a smaller scale. Instead, as defined here, pumped water storage takes advantage of stored potential energy and/or material by filling tanks and other storage devices with process or treated water. During peak usage, this water is available with no or minimal electricity, allowing for effective load shifting. An example system is shown in Fig. 7, where there is a need for process water and the capacity to store it.

While this strategy is not as popular as others, it is highly effective for applicable industrial processes. These solutions do not tend to be as scalable as a typical pumping system is much larger with demand reduction ranging from 850 to 2000 kW. Papers have reported a similar percentage in demand and cost reduction, ranging from 3 % to 5 %. However, because the infrastructure was already onsite, the payback periods were very low (Ashok and Banerjee, 2000; Chen et al., 2024).

Combining pumped water with renewable energies provides a way to convert non-dispatchable to dispatchable energy. By integrating water-pumping systems with PV energy generation, Chen et al. reduced the total electricity provided by the grid but increased the payback period from 0.7 to 10.4 (Chen et al., 2024). A study by Almehizia et al. showed that by combining electrochemical storage, pumped water storage, and renewable energy, the industrial facility achieved up to 50 % renewable energy penetration (Almehizia et al., 2019).

A summary of the data in Table 4 depicts the larger sizes of Pumped Water Storage. While the demand reduction is typically larger, the reduction percentage is smaller than other storage techniques. However, the payback periods are typically much smaller because the existing infrastructure is usually already in place. These examples also illustrate how pumped water storage plays a crucial role in renewable resource integration.

3.5. Other storage

Aside from the aforementioned energy storage techniques, many other energy storage techniques are more applicable in specific situations. One studied approach is the use of supercapacitors for energy storage. Supercapacitors are similar to electrochemical batteries but typically have higher discharge rates and smaller quantities of energy storage (Mensah-Darkwa et al., 2019). Elio et al. showed both the viability of this tactic and the dependence on profile type by comparing two profiles: a typical profile for a manufacturer and another for a wastewater treatment plant. The difference in peak shavings (4.5 vs. 2.1 % respectively) gives an example of the reliance of storage systems on profile type and shape. Ammonia plants offer another promising example of IDR by storing ammonia. Additionally, peak-time electricity demand was reduced by up to 7.4 % and costs by 1.4–4.0 % (Kelley et al., 2022a). Similar studies for a cement plant (storing raw mill) and a chlor-alkali plant (storing hydrogen from a fuel cell) continue the trend shown in the above storage sections. The range of demand shifting is between 2 % and 30 % but tends to be quite costly if the storage mechanism is not already available onsite (Summerbell et al., 2017; Wang et al., 2017).

While these technologies may be more site-specific, Table 5 highlights the ingenuity of implementing unique storage systems to shift demand. The table shows that typical demand reduction percentages are under 10 %, and payback periods can still be long if extra equipment is required. However, these ideas are valuable as they expand the scope to help unique or obscure facilities that may have unique opportunities, especially those with existing infrastructure for storage.

3.6. Summary of storage solutions

Storage technologies enable users to change the time of day electricity is bought from the grid, charging storage devices when electricity is cheap or demand is low to lower grid electricity dependency. Fig. 8 and Fig. 9 outline the percentage of load shifted, the payback period, and the impact of timing on the facility's energy use for the various storage solutions. The load shifted and payback periods are based on quantitative numbers from references, while the impact is set with a qualitative description of each storage solution. As mentioned throughout the section, the percentages in Fig. 8 show that achieving above 20 % reduction of peak demand using energy storage techniques is challenging. Technologies with shorter payback periods, such as TES or pumped water

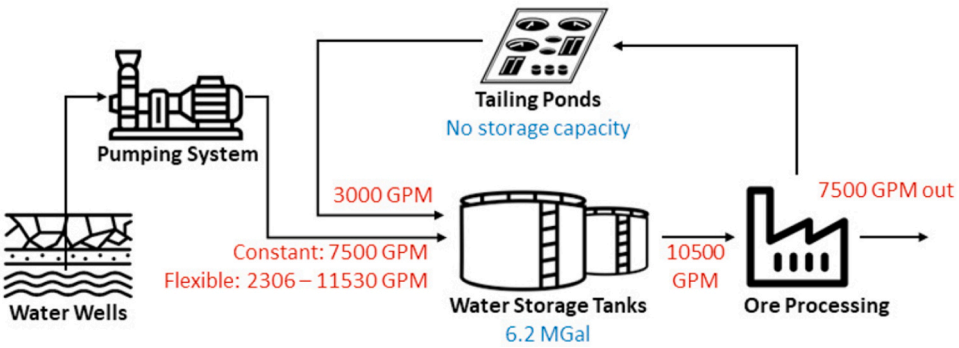


Fig. 7. Process Flow Diagram of a facility with the need and capability to store pumped water. Image acquired from Chen et al (Chen et al., 2022).

Table 4
Peak Reductions, Payback Periods, and Synergistic Technologies of Pumped Water Storage.

Paper Title	Demand Shifting Method	kW Shifted /Reduced	kWh Shifted /Reduced	Savings Mechanism	Payback Period (years)	Monetary Savings (per year)	Price
Largest Peak Reduction							
(Ashok and Banerjee, 2000)	Pumped Water Storage	**850** **3.6 %**		Energy costs, demand costs		$445,000 (₹39 Million) 3 %	
(Fekri Moghadam et al., 2015)	Pumped Water Storage, RTM	**70,000** **7.8 %**	140,000 4.5 %	Demand costs			
Price and Payback Period (for existing storage tanks)							
(Chen et al., 2024)	Pumped Water Storage, RTM	1,500, 4 %	None	Demand costs	**0.7**	$371,900, 2.3 %	**$300,000**
Pumped Water + Energy Generation							
(Chen et al., 2024) (Almehizia et al., 2019)	Hydro Pumping Storage, RTM, PV, TES	5, 50 %	4 %	Demand costs	10.4	4.7, 10 %	$8,000,000

Table 5
A Summary of Other Energy Storage Systems.

Paper Title	Demand Shifting Method	kW Shifted /Reduced	kWh Shifted /Reduced	Savings Mechanism	Payback Period (years)	Monetary Savings (per year)
(Elio et al., 2022)	Energy using a supercapacitor	**20** **4.5 %**	100	Energy costs, demand costs	41.2	$15,624
(Elio et al., 2022)	Energy using a supercapacitor	**20** **2.1 %**	100	Energy costs, demand costs	30.1	$21,396
(Kelley et al., 2022a)	Ammonia Storage, Real-Time Scheduling, Optimization	**5.4, 11.2** **3.57, 7.40 %**		Energy costs, demand costs		$2,624, 3,916 1.39, 3.98 %
(Summerbell et al., 2017)	Crushed Stone, Raw Meal, Scheduling Production	**1,000** **30 %**		Demand costs		$440,000 (£350,000) 4.2 %
(Wang et al., 2017)	Hydrogen from a Fuel Cell, Hybrid Generation		945	Demand savings, energy costs		$31,000

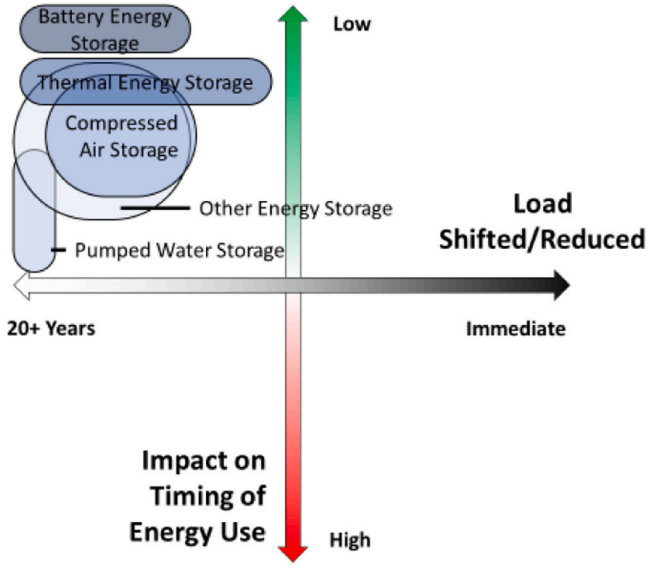


Fig. 8. Impact on timing of energy use vs. load shifted or reduced for storage solutions. Original image.

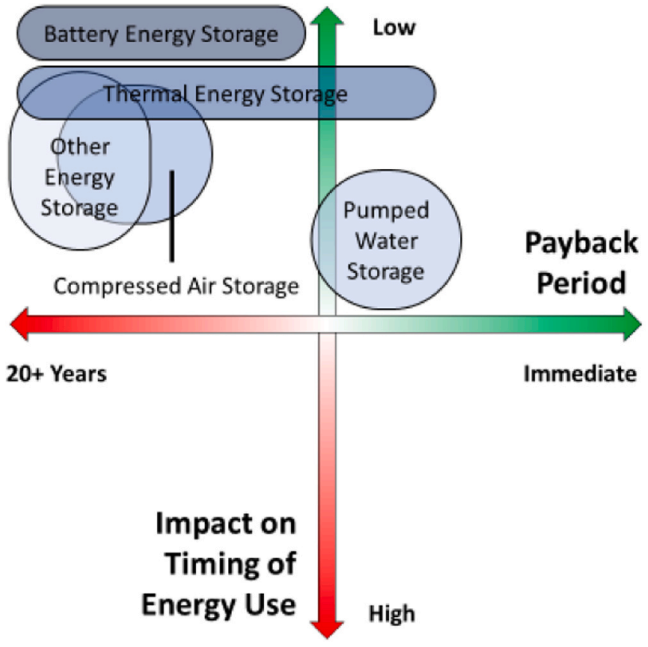


Fig. 9. Impact on timing of energy use vs. payback period for storage solutions. Original image.

storage, utilize existing on-site facilities and are typically not sized large enough to achieve high peak shavings. Technologies with longer payback periods, such as BESS and CAES, are too expensive and do not give adequate financial benefits for installing large systems, thus limiting their demand IDR potential. Considering energy storage as a whole, energy storage results in minimal impact on the timing of energy use by the facility, meaning that the facility can operate normally with few or insignificant changes. However, it comes with more capital costs, limiting its use in some situations.

4. Scheduling: planned decrease in energy usage

Scheduling is a popular solution for many industrial facilities seeking to lower demand peaks or reduce electricity costs. This section

overviews many techniques facilities use to decide how operations should run as a standing schedule to reduce electricity use and demand. These techniques include deciding when to run batch operations, how to cycle identical manufacturing lines, staggering startup of machinery, or when to turn machines off. While these solutions change how and when a manufacturer may operate, they do not typically consider real-time decision-making strategies.

4.1. Scheduling production

Scheduled production can be a powerful strategy in IDR, enabling qualifying facilities to schedule production so that energy is consumed during periods of low demand or high renewable energy availability. This approach focuses on shifting production processes to off-peak hours, though the total quantity of manufactured goods in a day, week, or month goes unchanged. This method reducing costs and improving energy efficiency without requiring significant capital investments or production capacity. A good illustration of this idea can be found in Fig. 10. By scheduling jobs away from peak TOU rates, as shown in case b, jobs can still be completed within the needed time, but attempt to avoid high electricity rates, depicted at the bottom of the figure.

Scheduling production is a straightforward way to manage electricity demand through operational changes rather than capital expenses. When implementing this method, peak demand savings of 6–20 % are common (Pablos et al., 2021; Kholerdi and Ghasemi-Marzbali, 2021). This method is also very scalable, meaning it can be used in both small and large facilities. A study by Zaidi et al. and a study by Reka and Ramesh both resulted in a similar range of demand savings (13–20 %) despite the wide range of kW usage: 25 kW and 4000 kW, respectively (Zaidi et al., 2020; Reka and Ramesh, 2016). The savings can also be noticeable and comparable to energy storage techniques. By implementing operational changes with few capital costs, savings of 10–15 % and above have been reported (Pablos et al., 2021; Zaidi et al., 2020; Lu et al., 2021; Zhao et al., 2014). While the payback periods are often not considered in these studies, it can be assumed they are much lower than those of energy storage devices. The tradeoff is that the facilities have a much more significant impact on when they are using electricity.

While most applications use demand reduction as the savings mechanism, some studies have shown the ability to participate in TOU rates or DRP. Air separation units are a good source of scheduled production due to their ability to store produced gas (Kelley et al., 2020; Tsay et al., 2021). These studies have reduced demand by 8.91 % during demand response calls. Furthermore, the approach has minimal process impact, with only a ~5 % reduction in production (with the added benefit of reducing carbon emissions), making it a cost-effective solution that avoids additional equipment costs (Kelley et al., 2022b, 2019). Similarly, flexible compressed air generation enables load shifting by controlling production systems and compressed air supply in tandem, aligning energy use with renewable energy availability. This system achieved a demand shift of 6.5 kW, representing a 22 % reduction in peak energy use, without affecting throughput (Beier et al., 2015). However, scheduling production is not always productive. Altering process timing can be productive at lowering peak electricity usage but can lead to a loss in production, as shown by Ameri Sianaki et al (Ameri Sianaki et al., 2018).

Scheduled production is sometimes paired with storage technologies. This results in two sources of peak reduction: a reduction due to

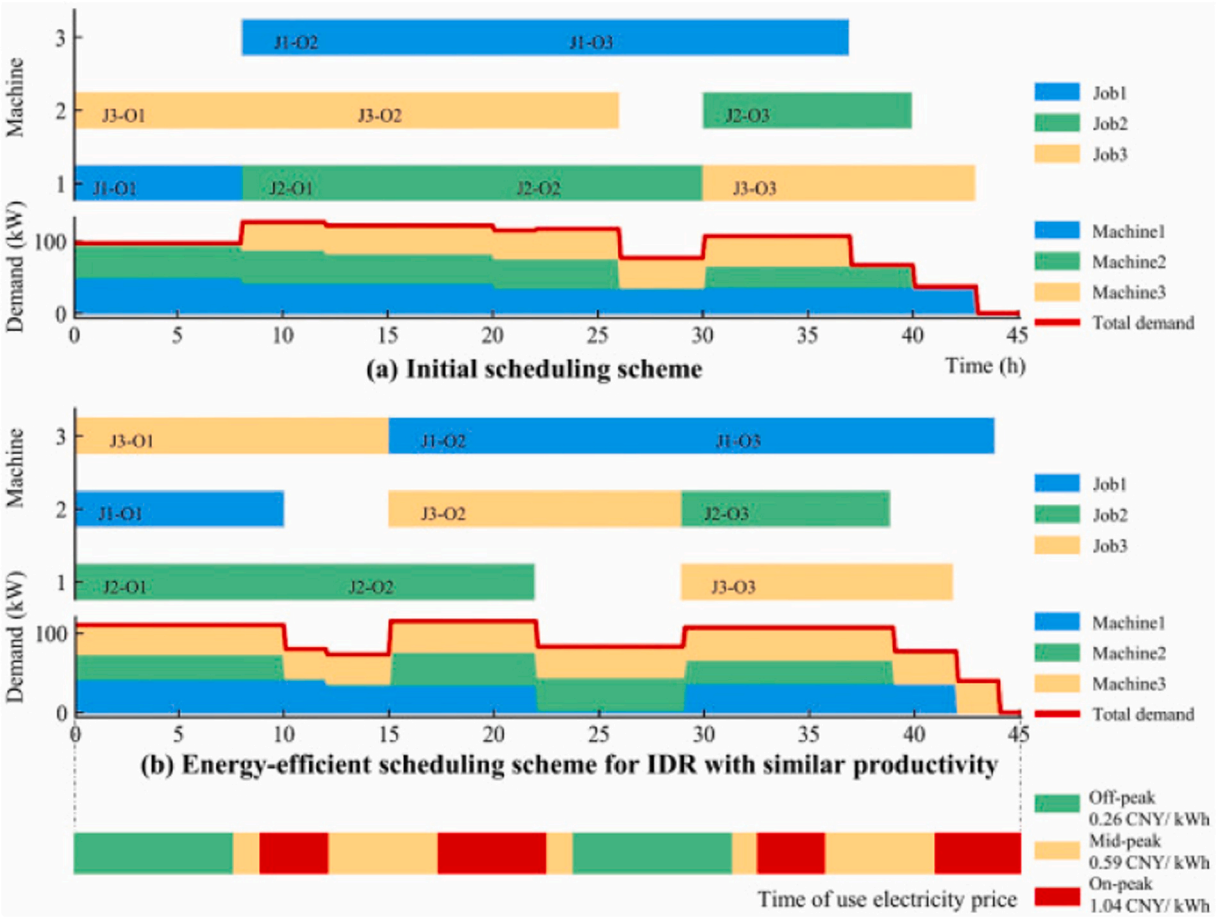


Fig. 10. Illustration of scheduled production, where the energy-efficient scheme shifts blocks of electricity usage to lower overall demand. Image acquired from Rui et al (Rui et al., 2024).

rescheduling manufacturing and a reduction due to the discharge of stored energy. An ideal situation is a partial reduction by both solutions, as shown by Yun et al. The approach reduced electricity costs by 15.1 % and peak demand by 31.8 % (Yun et al., 2022). This method is also productive for demand response programs (DRP). After scheduling production to have the smallest load factor, an onsite BESS can displace the rest of the load to respond to DR requests. A robust two-stage optimization framework conducted by Ruiz Duarte et al. aligned production schedules with renewable energy availability and achieved a 10 % cost reduction (Ruiz Duarte et al., 2020). Additionally, scheduling production in cement plants by shifting operations to low-cost or low-carbon periods reduced power consumption by approximately 1 MW or 30 %. This approach is made feasible by the ability to store excess material between batches (Summerbell et al., 2017). These and other storage opportunities outline the typical peak reduction of scheduled production and storage to be between 10 % and 30 % (Beier et al., 2015; Westberg et al., 2019).

While the above practices include moving blocks of energy usage, another opportunity is to simply lower energy usage through variable frequency drives (VFDs) and dynamic equipment control (DEC). A successful approach incorporates understanding which machinery can have a variable process rate and which cannot, as an example is shown in Table 6. Applied to a steel powder manufacturing facility, this method achieved a 9.8 % reduction in energy costs and shifted 5 % (150 kWh) of energy use to off-peak periods (Huang et al., 2017). When applied to a series of pumps (turning some off, leaving others on minimal flow rates), this can rise to an impressive 76 % peak reduction (1062 kW) (Machalek et al., 2020a). A similar approach, cycling production, can achieve comparable results. When multiple identical energy users have constant energy demands, selectively shutting some down while others remain operational can reduce peak loads. For example, in a steel powder manufacturing facility, this method reduced peak demand by 40 % of the original load (Huang et al., 2019).

One innovative approach is to combine scheduled production with energy generation. The effectiveness of this method results from energy being used when it is being generated rather than stored. This creates much shorter payback periods, not requiring the capital cost of an energy storage technology. Both dispatchable natural gas and non-dispatchable PV electricity can heighten the savings of systems integrating scheduled production, resulting in possible demand reductions of 10–38 % (Ruiz Duarte et al., 2020; Montuori et al., 2021) One study using this strategy even reduced energy use by 80 % during critical periods (Weitzel and Glock, 2019). These results are similar to the percentages when combining energy generation with energy storage.

Table 7 summarizes the ranges of peak reduction and savings from the gathered research. Because scheduling production is applicable in various industrial processes, it proves to be a very scalable solution for IDR. The data shows that it can result in considerable savings, both for total demand decreased and the percentage of peak demand. Scheduling production can result in large monetary savings while requiring little capital cost. By optimizing production timing, peak demand usage can be lowered, resulting in reduced electricity costs and increased grid stability.

Table 6
Example of Machinery with and without Variable Energy Demands.

Equipment Type	Energy Demand (kW)				
	Power 1	Power 2	Power 3	Power 4	Power 5
Evaporator	0	15	25	-	-
Crystallizer	0	60	-	-	-
Harvester	0	25	50	-	-
Crusher	0	15	20	-	-
Sieve	0	30	-	-	-
Dryer	0	20	30	40	55
Iodizer	0	10	-	-	-
Packager	0	6	10	-	-

Table 7
Peak Reductions, Size, Savings, and Synergistic Technologies of Scheduled Production from Select Studies.

Papers used	(Weitzel and Glock, 2019; Yun et al., 2022; Ruiz Duarte et al., 2020; Huang et al., 2017; Pei et al., 2019; Beier et al., 2015; Summerbell et al., 2017; Rui et al., 2024; Pablos et al., 2021; Kholerdi and Ghasemi-Marzbali, 2021; Zaidi et al., 2020; Reka and Ramesh, 2016; Lu et al., 2021; Ameri Sianaki et al., 2018; Westberg et al., 2019; Machalek et al., 2020a; Huang et al., 2019; Montuori et al., 2021)	
	Median	IQR
Demand Shifted or Reduced (kW)	175	45 – 5,750
Percent of Demand Shifted or Reduced (%)	22	10.8 – 35
Energy Shifted or Reduced (kWh)	109	50 – 321
Percent of Energy Shifted or Reduced (%)	34	11.8 – 37.5
Monetary Savings (USD per year)	12,900	9,440 – 21,400
Percent of Monetary Savings (%)	10	5.9 – 18.5

4.2. Cycling production

Cycling production involves selectively turning off or reducing the activity of duplicate machines during periods of high electricity demand. Often, industrial facilities will have multiple manufacturing lines or equipment that are identical or similar. Only some lines or machines must be turned off to reduce the peak to the average electricity load and achieve demand savings. This approach helps balance the need to reduce peak demand and keep production flowing. It is different than scheduled production in that does not completely shut down a manufacturing step, but only decreases throughput for a certain amount of time. Demonstrated for a food production facility, Angizeh et al. cycled equipment to schedule appropriate production at off-peak times. The approach shifted approximately 20 % of the load during high-demand periods, shown in Fig. 11 (Angizeh et al., 2020). A similar approach was shown to work in a metal casting process, lowering the peak demand by 5–10 % (Ramin et al., 2018). These examples align with others to show that this method can shift a noticeable amount of demand, though not as much as scheduling production (Westberg et al., 2019; Batista Abikarram et al., 2019). The typical demand reduction lands in the range of 5–20 %.

Combining cycling production with other strategies is an effective way to increase the peak reduction past 20 %. Optimization techniques and scheduling production can see a sharp decrease in demand, reaching up to 70–80 % for short periods. This reduction is made possible by cycling some while scheduling others for off-peak hours. However, these reductions are typically for shorter periods to take advantage of times with high electricity prices. The high peak reduction is canceled out by the short reduction period to produce similar cost savings: 10–20 % (Huang et al., 2019; Desta et al., 2018; Lu et al., 2020).

Table 8 highlights the findings from select facilities that have implemented cycling. The peak reduction percentages lay around 20 %, similar to the results of storage technologies. However, the capital costs of storage technologies have been replaced with the need for flexibility, allowing for better savings and payback periods. The demand reduction is less than the possible range from scheduling production, likely because of the need to keep some production lines active. A summary of the process impact is included in the table to show the typical operational changes required.

4.3. Scheduling start-up periods

Scheduling the start-up of industrial machinery is a key IDR strategy that optimizes energy consumption by strategically dictating when

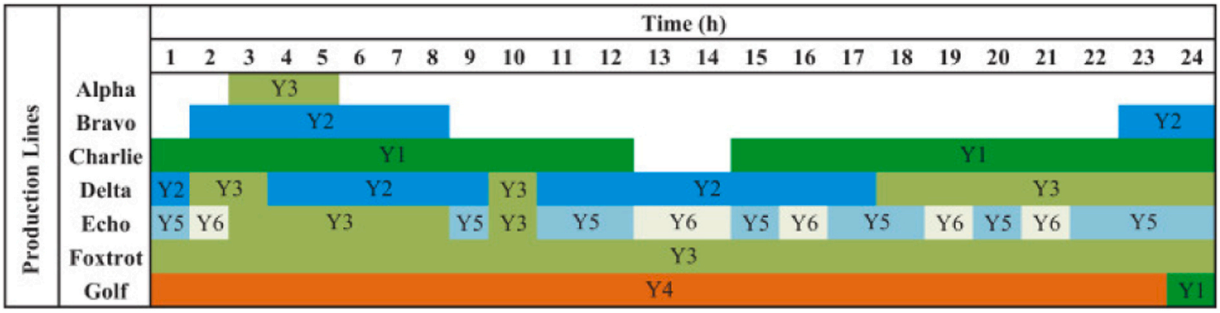


Fig. 11. An example of cycling production, as seven manufacturing lines are optimized to keep electricity use low between the hours of 12 and 14. Image acquired from Angizeh et al (Angizeh et al., 2020).

Table 8
Peak Reductions, Savings, and Synergistic Technologies of Cycling Production from Select Studies.

Papers used	(Westberg et al., 2019; Huang et al., 2019; Angizeh et al., 2020; Ramin et al., 2018; Batista Abikarram et al., 2019; Desta et al., 2018; Lu et al., 2020)	
	Median	IQR
Demand Shifted or Reduced (kW)	50	34 – 100
Percent of Demand Shifted or Reduced (%)	26.5	13.75 – 60.25
Monetary Savings (USD per year)	5,500	2,700 – 8,700
Percent of Monetary Savings (%)	12.5	10.1 – 21.5

equipment is activated. The highest demand for most machinery may occur when it is first powered on (U.S. Department of Energy, 2012). Significant peak demand can occur at many manufacturers when many pieces of machinery are started at once at the beginning of shifts or when a particular manufacturing step is needed, as an example illustrates in Fig. 12. While this example uses HVAC to depict a possible current and proposed solution, this can be applied to a variety of machinery in industrial settings such as boilers, chillers, compressors, or any large equipment that have multiple units onsite. Scheduled start-up can be implemented manually or through the use of self-scheduling systems. However, this method is only effective when the peak demand occurs when multiple machines are turned on simultaneously, most common in industries that shutdown overnight or over weekends. This method is less effective if the peak demand occurs at a different time.

When a demand peak can be tied to the powering up of multiple pieces of machinery, demand savings of 13–40 % can be seen (Golmohamadi et al., 2020; Brimley et al., 2019). When considering dynamic electricity pricing, this method reduced demand by 69 % under critical peak pricing by strategically timing equipment activation based on real-time electricity prices. Although shifting start-up times extended production duration, this trade-off effectively reduced demand and energy costs (Gong et al., 2015).

Combining CHP systems with smart scheduling represents another innovative method for managing start-up operations due to the large size of CHP systems. In a study by Henning et al., a microturbine is utilized in a discrete manufacturing facility to simultaneously meet thermal and electrical demands. By scheduling equipment start-up in coordination with CHP operation, the facility increased from a 58 % reduction from using the CHP system to a 65–70 % reduction (917–997 kW) and cost savings of $1200 per day (Henning et al., 2019).

Table 9 outlines the results of demand reduction and savings. Though scheduled startup is a smaller operational change than scheduling or cycling, it has been shown to increase the peak reduction percentage, reaching at times 20–30 %. These were accompanied by cost savings of 20–40 %. Though this tactic is applicable only in some scenarios, it can effectively lower demand costs with little to no operational or capital costs or impact on productivity in applicable scenarios.

4.4. *Curtailment*

Curtailment is the strategic shutdown or temporary deactivation of

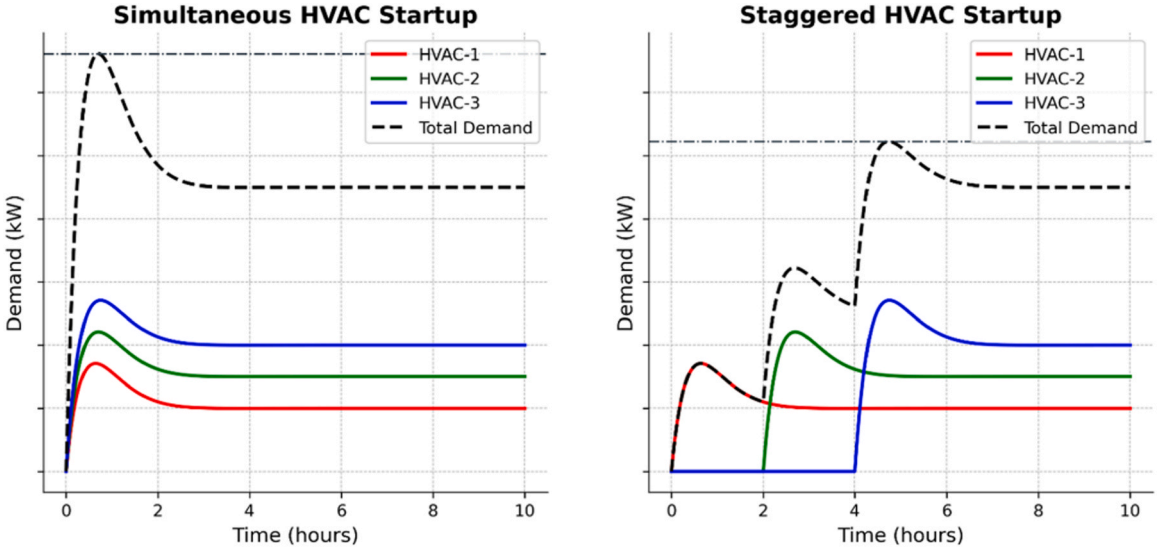


Fig. 12. An example of how scheduled startup can reduce the peak usage, often called staggered starts. Original image.

Table 9
Peak Reductions, Saving, and Synergistic Technologies of Scheduled Startup from Select Studies.

Papers used	(Golmohamadi et al., 2020; Brimley et al., 2019; Gong et al., 2015; Henning et al., 2019)	
	Median	**IQR**
Demand Shifted or Reduced (kW)	817	418 – 957
Percent of Demand Shifted or Reduced (%)	30	25.5 – 58
Energy Shifted or Reduced (kWh)	16,200	15,400 – 17,000
Percent of Energy Shifted or Reduced (%)	75[1]	
Monetary Savings (USD per year)	13,200	9,000 – 14,400
Percent of Monetary Savings (%)	40	31 – 54.5

[1]This 75 % is the reduction of one piece of equipment, not the entire industrial demand

machinery during periods of high energy demand. While scheduling or cycling production may move manufacturing from one time of the day or week to another, energy that is curtailed will not increase later energy consumption. This is achieved by relaxing setpoints, turning off unneeded equipment or possibly lowering production throughput. While this method is typically more challenging to implement in industrial facilities, it can be highly effective in certain situations. This method can result in savings for both demand and energy as it simply turns the power off and does not shift energy use to other times of the day.

Curtailment proves to be effective with air separation. Data-driven approaches can save money by dynamically aligning production rates with electricity price shifts. Adjusting curtailment to a day-ahead pricing schedule has achieved a 2.7–3 % decrease in energy cost despite a maximum of 85 % decrease in demand used by the air separation units during high-price times (Tsay et al., 2019a; Pattison et al., 2016). However, when the same process is used but applied to a real-time pricing market, the savings increased to 13 % with a similar decrease in peak demand reduction (Tsay et al., 2019a). When comparing the energy reduction to the entire site's energy usage, a 20 % decrease in facility-wide peak reduction was achieved (Kelley et al., 2018). A comprehensive view of these air separation units shows that the energy price can decrease by a maximum of 46 % during times of power line congestion (Baldea, 2020).

This method is also very successful in batch manufacturing as batches can easily be scheduled to run during off-peak times, curtailing the energy needed for the whole process. A study done for a bakery showed that the facility reduced energy peaks by 11 % by turning off parts of the process when the machinery requiring supply was also turned off (Westberg et al., 2019). Another easy solution is to decrease setpoints for HVAC systems that do not require strict temperature controls. A study by Brundage et al. demonstrated how dynamically adjusting HVAC setpoints during peak demand periods reduced overall energy costs by over 5 % while maintaining operational efficiency and worker comfort (Brundage et al., 2014).

Table 10 outlines the summary of curtailment from select studies. The demand reduction percentages are similar to the other three scheduling techniques. However, most of the applications for curtailment are economically feasible due to the use of real-time markets. Therefore, the savings percentages are not as large as in other sections. Curtailment is one solution for IDR but must be combined with other solutions to be highly effective for demand or cost savings.

Table 10
Peak Reductions, Saving, and Synergistic Technologies of Scheduled Curtailment from Select Studies.

Papers used	(Diekerhof et al., 2016; Westberg et al., 2019; Henning et al., 2019; Tsay et al., 2019a; Pattison et al., 2016; Kelley et al., 2018; Baldea, 2020)	
	Median	IQR
Demand Shifted or Reduced (kW)	60	19 – 180
Percent of Demand Shifted or Reduced (%)	11	10 – 20
Energy Shifted or Reduced (kWh)	7,480	3,750 – 11,200
Percent of Energy Shifted or Reduced (%)	60.5[1]	
Monetary Savings (USD per year)	8,050	3,510 – 13,000
Percent of Monetary Savings (%)	4.15	2.8 – 11.1

[1]This 75 % is the reduction of one piece of equipment, not the entire industrial demand

4.5. Summary of scheduling techniques

Some scheduling techniques allow less machinery to operate during peak demand, higher electricity rates, or higher demand rates. Others ensure that peaks from machinery do not coincide or allow for the energy demand of processes to be turned down. These solutions work well on their own or in combination with other technologies. Fig. 13 and Fig. 14 outline the percentage of load shifted, the payback period, and the impact of timing on the facility's energy use for the various storage solutions. A typical demand reduction for scheduled production can be fairly high, reaching around 60–70 %. Most solutions have low capital costs, allowing for reasonably short payback periods. The tradeoff for these solutions is that they result in a more significant change in the timing of energy use by the facility, changing how and/or when the manufacturing lines operate. Unlike storage solutions that could be used at any industrial facility, this change in timing may not be possible or preferred in every situation.

5. Operational flexibility: real-time energy shifting

Operational flexibility solutions resemble scheduled production but rely on real-time responses to facility or market inputs. A predefined savings mechanism adjusts energy use only under certain conditions, enabling facilities to manage peak demand, optimize real-time pricing, and respond to demand response requests.

5.1. Real-time scheduling

Real-time scheduling (RTS) involves dynamic adjustments to energy use, made in real-time based on electricity price fluctuations or demand response programs. This approach enhances flexibility, allowing

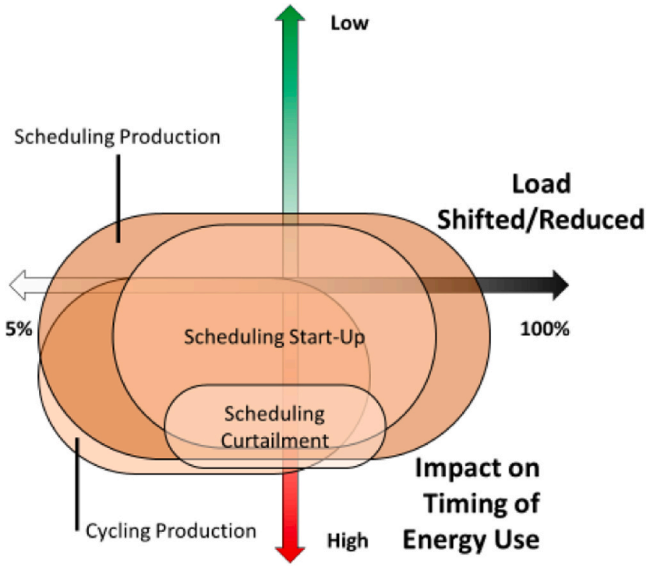


Fig. 13. Impact on timing of energy use vs. load shifted or reduced for scheduling techniques. Original image.

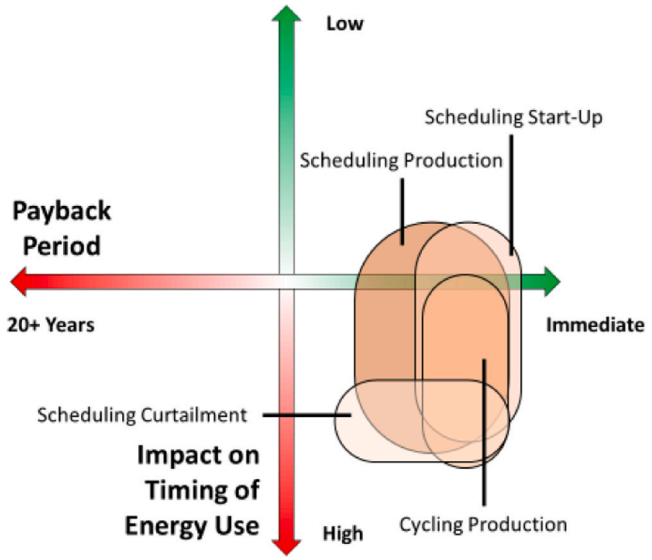


Fig. 14. Impact on timing of energy use vs. payback period for scheduling techniques. Original image.

facilities to optimize energy costs without compromising production. This method must be combined with real-time monitoring or predictive modeling to make real-time adjustments based on the energy market. These methods typically have a pre-established logic flow and some system of control software for the facility. Because RTS happens in real-time, it is not typically controlled manually (Zhou et al., 2020).

Industrial processes relying on electrochemical processes, like chlor-alkali production, showcase significant potential for RTS. For example, a dynamic model tailored for chlor-alkali plants enables real-time adjustments based on current electricity price by modulating current density, reducing cell power demand during peak price periods. This strategy achieved 21–29 % demand reduction and 7.3 % energy cost savings over 72 h (Otashu and Baldea, 2019; Simkoff and Baldea, 2020). Despite the lower peak savings, the electricity cost savings increased to 9.0 %. Using a state-task network model to shift activities according to changing rates in a flour mill, one study achieved a 38 % reduction in energy costs while maintaining production output (Zaidi et al., 2020).

RTS is commonly combined with power storage and generation to enhance peak shaving. A steel powder facility using RTS and DEC achieved a 33 % peak reduction and 9.9 % cost savings, increasing to 57 % and 17 % when integrated with TES. However, RTS alone reduced peak electricity demand by 3.57–7.4 % and operational costs by up to 3.98 % by storing excess ammonia during off-peak hours, showing the effectiveness of RTS alone (Kelley et al., 2022a). In CHP systems, RTS optimized heat and power generation, cutting daily energy costs by 4.3 % while balancing electricity and natural gas prices (Alipour et al., 2019, 2014). RTS also aids demand management in multi-user systems, such as aluminum, steel, and cement production. In a microgrid with predictive analytics and wind power, RTS reduced overall load by 20 %, peak load by 31.67 %, and increased revenue by 41.32 % through energy market participation (Gerami et al., 2021).

Finally, facilities can benefit from RTS by synchronizing curtailment with real-time market prices. With RTS, there is no need to regularly curtail operations, only during times of high pricing or during demand response requests. Air separation units reduced energy costs by 13 % in real-time markets, showcasing the importance of curtailment and process integration in optimizing energy use (Tsay et al., 2019a).

A summary of the tabulated data in Table 11 shows the typical demand reduction to be around 19–33 %, with higher percentages possible for short durations when electricity prices are high. These were all accomplished on smaller systems; this suggests that RTS is not as scalable as other solutions. Because of the short duration of these applications, the cost savings are smaller, though still near 10 %.

Table 11
Peak Reductions, Saving, and Synergistic Technologies of Real-Time Scheduling from Select Studies.

Papers used	(Huang et al., 2017; Kelley et al., 2022a; Zaidi et al., 2020; Otashu and Baldea, 2019; Simkoff and Baldea, 2020; Alipour et al., 2019, 2014; Gerami et al., 2021)	
	Median	IQR
Demand Shifted or Reduced (kW)	130	70 – 400
Percent of Demand Shifted or Reduced (%)	20.5	12.3 – 32
Energy Shifted or Reduced (kWh)	195	145 – 250
Percent of Energy Shifted or Reduced (%)	8	6.5 – 13.5
Monetary Savings (USD per year)	5,480	3,820 – 19,100
Percent of Monetary Savings (%)	9.9	4.14 – 15

5.2. *Dynamic Equipment control*

Dynamic Equipment Control (DEC) enables machinery to adjust its energy use in real-time, providing flexibility for industrial facilities to participate in real-time programs such as real-time pricing and demand response programs. This dynamic control is typically done by VFDs, which can lower the energy use of a motor or other mechanical device. VFDs often result in low payback periods before IDR applications due to the ability to decrease the amount of electricity needed for a piece of machinery, all at a lower capital cost (Miller et al., 2012; Kapp et al., 2024; Dutta et al., 2020). Adding savings from demand shifting increases their economic outlook.

A typical application uses VFDs to adjust material flowrates through certain parts of the process. They can result in significant peak shavings, momentarily avoid high electricity prices, or participate in IDR. The range of reports shown below is between 14 % and 52 %. By dynamically controlling the power supply to air dryers and distribution, a Spanish cured ham factory reduced peak power demand by 50 % and saved 5 % on electricity costs annually without compromising product quality (Alcázar-Ortega et al., 2012). Adjustments in kiln rotation, stone feed, and coal feed rates in a rotary kiln enabled power reductions of up to 1.4 MW. This approach reduced poor product quality to zero and cut economic penalties for short-term grid responses by 30–70 %, depending on the daily load profile (Machalek and Powell, 2019b). Similar results were seen in an electrolytic aluminum facility, reducing power use by up to 25 % during peak demand periods (Gong et al., 2021).

A popular method is to combine DEC with scheduled production. This allows schedulable tasks to be moved to off-peak hours and remaining non-schedulable tasks to be ramped down using DEC. When implementing these in a mining facility, Machalek et al. reported that the demand peak reduction increased from 14 % to 76 % when VFDs were used to control the non-schedulable use of pumps (Machalek et al., 2020a). A steel powder manufacturer reduced peak demand by 33 % using DEC control. Switching from day-ahead to hour-ahead pricing did not improve peak shaving but increased cost savings from 3.5 % to 9.9 % (Huang et al., 2017). However, DEC is not successful for every implementation. This method optimized energy allocation in a press-shop factory, resulting in peak demand and cost savings of 32 %. However, these savings could not replace the lost income due to decreased production (Ameri Sianaki et al., 2018).

Another successful strategy is to integrate DEC with energy storage and generation. The variable use of energy made possible by DEC allows energy storage and generation to operate with the same flexibility. The same steel powder manufacturer mentioned above increased peak shaving from 33 % to 57 % with the addition of TES, increasing the cost savings from 10 % to 17 % (Huang et al., 2017). While this is a good improvement, it is important to note the capability of DEC alone without the additional cost of TES. Adding a BESS to DEC reduced peak demand

by 15 % in wastewater treatment plants, rising to 30 % with a CHP system—again highlighting DEC's impact before costly additions (Musabandesu and Loge, 2021). DEC can also optimize natural gas generation, adjusting output based on real-time pricing. This can provide additional peak demand savings due to the increase in flexibility, which has resulted in a 75–78 % peak reduction with considerable cost savings of 30–70 % (Seo et al., 2020; Dababneh and Li, 2019).

These studies illustrate the versatility and efficacy of DEC in IDR programs, with applications ranging from cooling systems to energy-intensive manufacturing processes. Table 12 depicts a higher demand reduction than RTS, typically 30–70 %. The cost savings are also larger due to the ability of VFDs to save demand and energy costs. Because VFDs also require some capital cost, this lengthens the payback period, placing DEC in a unique situation requiring both moderate capital costs and high flexibility of energy use.

5.3. Summary of Operational Flexibility

Fig. 15 and Fig. 16 outline the percentage of load shifted, the payback period, and the impact of timing on the facility's energy use for the two operational flexibility solutions. These figures show the typical lower payback periods but high energy usage impacts of these technologies. This high impact stems from the ability/need to reduce electricity usage in real-time. This results in a significant impact on the timing of energy use by the facility. A typical demand reduction for storage solutions is 25–45 %, with a maximum range of 5–75 %. Because operational flexibility allows for new savings mechanisms such as TOU markets and DRP without notable capital costs, the paybacks of these technologies are typically lower.

6. On-Site Energy Generation

Onsite energy generation is gaining momentum in industrial settings, especially those with large electricity loads. While the primary objective is typically to displace energy usage and not peak demand, both must be addressed if smooth integration with the electrical grid is to be achieved. In addition, considering peak demand charges can increase the financial incentives driving implementation. Even when a technology is oversized, the excess electricity can often be sold back to the grid. While the presented technologies in this report are not exhaustive, they represent the most common solutions present in current research: CHP, natural gas turbines, solar, wind, and hybrid solar-wind configurations. Some viable options not as well researched for IDR applications include geothermal, biomass, coal, and hydrogen.

Table 12
Peak Reductions, Saving, and Synergistic Technologies of Dynamic Equipment Control from Select Studies.

Papers used	(Musabandesu and Loge, 2021; Huang et al., 2017; Machalek et al., 2020a; Montuori et al., 2021; Alcázar-Ortega et al., 2012; Machalek and Powell, 2019b; Gong et al., 2021; Seo et al., 2020; Dababneh and Li, 2019)	
	Median	IQR
Demand Shifted or Reduced (kW)	500	120 – 1,060
Percent of Demand Shifted or Reduced (%)	33	26.3 – 66.8
Energy Shifted or Reduced (kWh)	370	150 – 1,140
Percent of Energy Shifted or Reduced (%)	5	3.8 – 8
Monetary Savings (USD per year)	13,000	8,000 – 23,000
Percent of Monetary Savings (%)	17	7.5 – 31

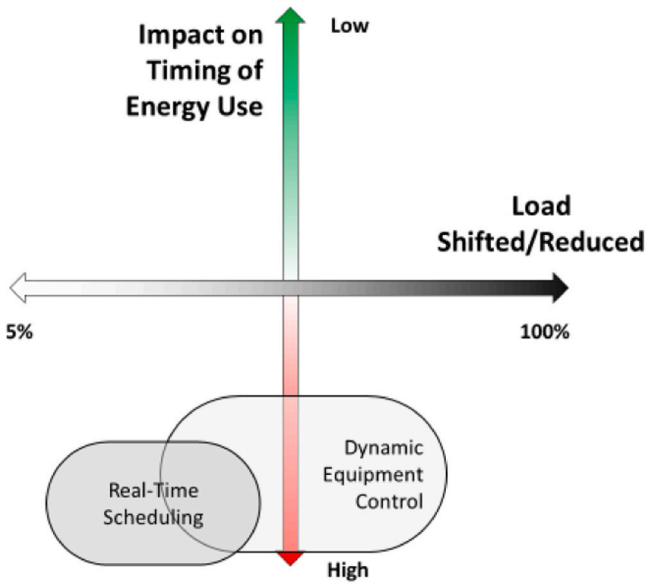


Fig. 15. Impact on timing of energy use vs. load shifted or reduced for operational flexibility techniques. Original image.

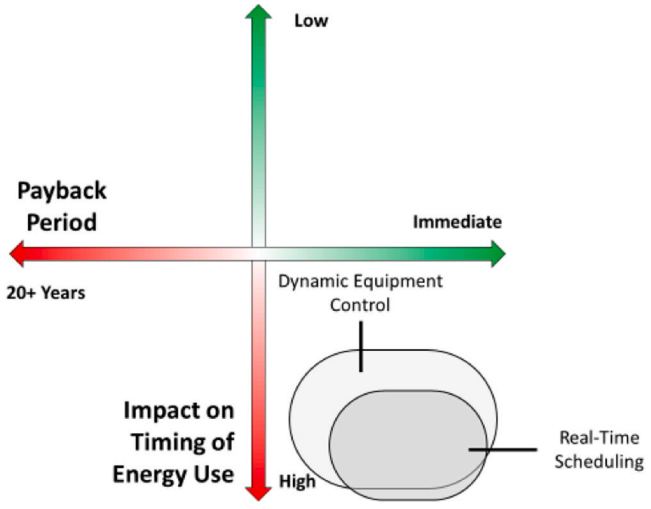


Fig. 16. Impact on timing of energy use vs. payback period for operational flexibility techniques. Original image.

6.1. Carbon Intensive Energies

6.1.1. Combined Heat and Power

Combined Heat and Power (CHP) systems represent a highly effective strategy for managing energy demand in industrial facilities by producing electricity and utilizing waste heat for thermal needs. By simultaneously addressing heat and electricity generation, CHP systems lower peak electricity demand, reduce greenhouse gas emissions, and improve overall efficiency. Because of the decreased energy loss due to heat and transmission, CHP systems can achieve efficiencies of 75–85 %, compared to a typical 36–60 % from off-site electricity (Overview of CHP Technologies, 2017; Elmer et al., 2015; Maghanki et al., 2013). Fig. 17 and Fig. 18 both show how heat and electricity are addressed, resulting in a higher efficiency. Facilities with large enough heat and electricity loads can achieve 6–8-year payback periods, which may be reduced to 2–3 years when incentives are included (Flexible Combined Heat and Power CHP Systems, 2018; Zhang et al., 2016b; Yin et al., 2021).

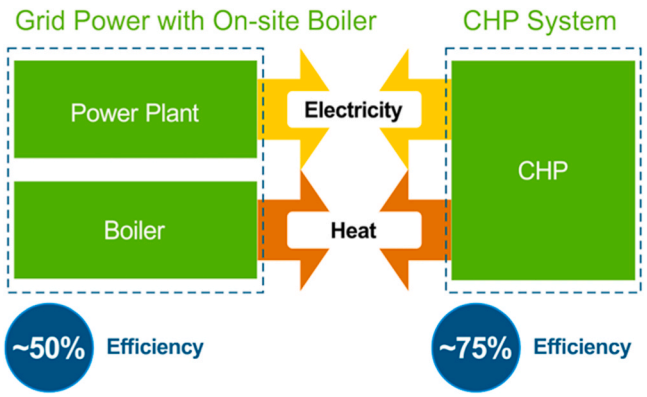


Fig. 17. A diagram depicting how CHP can address both heating and electricity needs in a more efficient way. Image acquired from the U.S. Department of Energy (Overview of CHP Technologies, 2017).

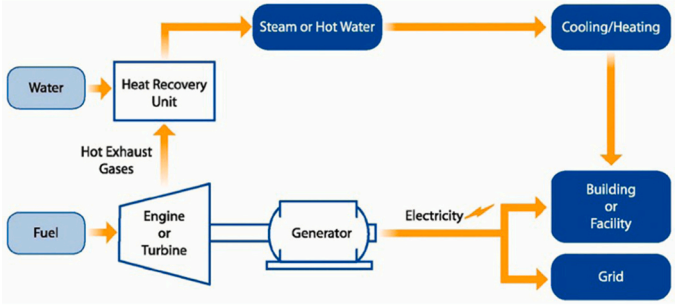


Fig. 18. A flow diagram depicting a typical CHP system. Image acquired from the U.S. Department of Energy (Overview of CHP Technologies, 2017).

Although CHP systems are not typically installed for DR purposes, there can be passive or active DR effects and savings. Demand savings can add savings that counter the increased cost due to natural gas use. Machalek et al. demonstrated this by displacing 73 % of the peak electricity use and 88 % of the energy use for a CHP user. This study resulted in the typical 8-year payback (Machalek et al., 2020b). While the peak reduction does not always need to be a large percentage, it can still displace megawatts of energy, as shown by Jiang et al., saving 40 % of the energy costs (Jiang et al., 2019). When the savings resulting from IDR are incorporated, the payback period can decrease to 3–5 years (Chiaroni et al., 2016; Celma et al., 2013).

Curtailment and scheduled production can enhance CHP systems by aligning energy generation with demand peaks. Using only CHP, a discrete manufacturing facility integrating a microturbine achieved 817 kW or 58 % reduction during peak times. When combined with scheduling, curtailment, or both methods, that reduction increased to 65–70 %, leading to an increase of 9 % in cost savings (Henning et al., 2019). CHP systems can also be integrated with real-time response mechanisms to increase savings. A wastewater treatment plant utilized CHP alongside battery storage and flow optimization to reduce demand by 30 % (500 kW) (Musabandesu and Loge, 2021). Exhibiting the common practice of oversizing a system, a battery manufacturing plant generated 150–163 % of its total peak (this was combined with other storage/generation techniques, mainly CAES, TES, and hybrid renewable generation) (Pei et al., 2019).

With 59 % of the energy provided to the electric power sector being lost, CHP systems can help reduce grid load due to their higher efficiencies and short payback periods (U.S. energy consumption by source and sector, 2023). A summary of the tabulated data is located in Table 13. The table depicts the first use of a technology capable of exceeding the peak demand usage, generating up to 163 % of the peak.

Table 13
Peak Reductions, Saving, and Synergistic Technologies of Combined Heat and Power Systems from Select Studies.

Papers used	(Jiang et al., 2019; Musabandesu and Loge, 2021; Pei et al., 2019; Henning et al., 2019; Machalek et al., 2020b)	
	Median	IQR
Demand Shifted or Reduced (kW)	957	817 – 6,190
Percent of Demand Shifted or Reduced (%)	61.5[1]	58 – 92.3
Energy Shifted or Reduced (kWh)	15,400	12,500 – 17,200
Percent of Energy Shifted or Reduced (%)	72	7.5 – 82
Total Energy Reduction (kWh)	6,000	3,000 – 55,000
Percent of Total Energy Reduction (%)	20	15 – 30
Payback Period (years)	4.3	3.45 – 5.75
Monetary Savings (USD per year)	13,200	13,000 – 14,400
Percent of Monetary Savings (%)	25.5	10 – 40
CHP Size (kW)	1,000	100 – 10,500

[1]This statistic is increased by (Pei et al., 2019) where energy was sold back to the grid and resulted in > 100 % demand reduction

Other systems may not exceed the demand need but can lower it significantly (30–73 %). The large monetary savings support the findings of shorter payback periods mentioned above.

6.1.2. Fuel-Based Power Generation

Electricity generators using natural gas as fuel are another technology that is not as popular but still widely used. These generators play a pivotal role in enhancing IDR, providing a reliable energy source to manage fluctuations in electricity demand and prices. Unlike CHP systems, these setups often focus solely on electricity generation without co-generation, leveraging demand-shifting strategies to optimize costs and energy use (Zhu et al., 2021; Santos and Uturbey, 2018; Hachem et al., 2022). A typical generator is depicted in Fig. 19.

The main success of natural gas generators for industrial applications typically comes from the ability to take advantage of both electrical and natural gas arbitrage. This often leads to a decrease in peak energy usage as well as a decrease in natural gas usage. These dual benefits can be seen in a study by Dzyuba et al., where they achieved a 33 % reduction in peak demand and reduced natural gas usage by 10 % (Dzyuba et al., 2023). Adding intelligent operations like predictive analytics, Cui et al. exhibited a similar decrease in peak demand of 13–73 %. This wide range of decreases depended on the day's load curve, exhibiting the reliance of IDR success on the shape of an industrial facility's load. While the natural gas usage increased in this case, there was still a net savings of 31 % of the total utility cost (Cui et al., 2016). Combinations of electricity generation from natural gas and scheduling or real-time monitoring have also resulted in the same range of peak reduction - 38–68 % - in two additional studies and showed an impressive 5.23-year

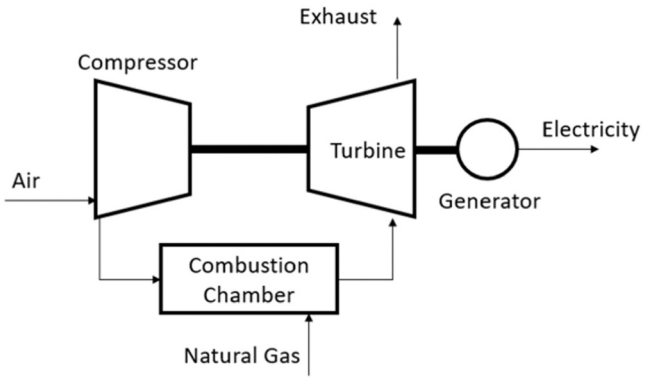


Fig. 19. Flow diagram depicting a typical natural gas generator, where natural gas combustion occurs in the combustion chamber to power the turbine, generating electricity. Original image.

payback period (Montuori et al., 2021; Zhang et al., 2016c).

Because of the ability to increase or decrease generation, natural gas generators can be successfully combined with operational flexibility methods. A typical range for this method is 75–78 % peak reduction, depending on the size of the natural gas turbine (Seo et al., 2020; Dababneh and Li, 2019). Incorporating real-time pricing models allows facilities to adjust natural gas generation dynamically, which reduced peak demand and cost by 19 % and 4.3 %, respectively, in one study (Alipour et al., 2019).

Because fuel-based power generators are typically smaller than CHP, the peak reduction is also smaller, as shown in Table 14. However, it is still very high compared to the other sections of this paper, with an upwards of 78 % reduction. Also unlike other sections of this paper, each of the papers cited contained cost savings resulting from electricity and natural gas arbitrage instead of using demand reduction. While electricity is typically the more challenging utility to match demand, electricity from natural gas can help address price and supply fluctuations in electricity and natural gas, creating multiple income sources to help fund this technology.

6.2. Renewable Energies

6.2.1. Solar

The integration of solar photovoltaic (PV) systems has emerged as a widely used renewable energy source in the industrial sector. However, for IDR applications, it is hard for PV alone to make an impact. For PV to be effective, energy generation and demand must have a coincident peak, or energy must be generated at the same time the facilities are experiencing peak demand (Fleschutz et al., 2022). In most situations, this is not likely. For example, a study by Chen et al. led to only a 0.2 % reduction in peak demand, a 3.5 % reduction in energy usage from the grid, and an extended 28.4-year payback (Chen et al., 2024). For this reason, PV is often paired with storage devices, as an example depicts in Fig. 20. By coupling these two, industrial facilities can reduce peak demand and participate in DRP (Carrillo et al., 2019; Huangfu et al., 2023; Zhu et al., 2016). When a BESS is added to the PV array in the study by Chen et al. mentioned above, the peak and cost savings increase, leading to a 17-year payback (Chen et al., 2024).

The combination of PV and BESS proves to be a very successful approach, requiring little to no energy adjustment from the facility. However, the payback is often very long because of the considerable capital cost for both solar and BESS. For instance, a microbrewery case study in South Africa highlighted the benefits of coupling PV systems with battery storage under DR conditions. Using a small 4.34 kW PV system and a 9.6 kWh lithium-ion battery, the brewery achieved a 58 % cost savings with a payback period of 13.8 years (Kusakana, 2020). In another study, which presented a dual-layer model for energy storage dispatching, the savings resulted in peak shaving of 40 % (Nan et al., 2021). Combining PV with TES typically results in similar payback periods (>15 years) due to the similar quantity of energy generated and reduction of peak demand (Saffari et al., 2018; Arteconi et al., 2017).

Table 14
Peak Reductions, Saving, and Synergistic Technologies of Fuel-Based Energy Generation Systems from Select Studies.

Papers used	(Montuori et al., 2021; Alipour et al., 2019; Seo et al., 2020; Dababneh and Li, 2019; Dzyuba et al., 2023; Cui et al., 2016; Zhang et al., 2016c)	
	Median	IQR
Demand Shifted or Reduced (kW)	5,970	2,400 – 17,297
Percent of Demand Shifted or Reduced (%)	53	29.5 – 73.5
Energy Shifted or Reduced (kWh)	4,150	1,890 – 29,100
Percent of Energy Shifted or Reduced (%)	27.5	15.8 – 39
Natural Gas Reduction (MMBtu/hr)	35.7	24 – 71.4
Percent of Natural Gas Reduction (%)	16.5	10.8 – 71.1
Monetary Savings (USD per year)	13,500	6,260 – 19,900
Percent of Monetary Savings (%)	18	4.3 – 31

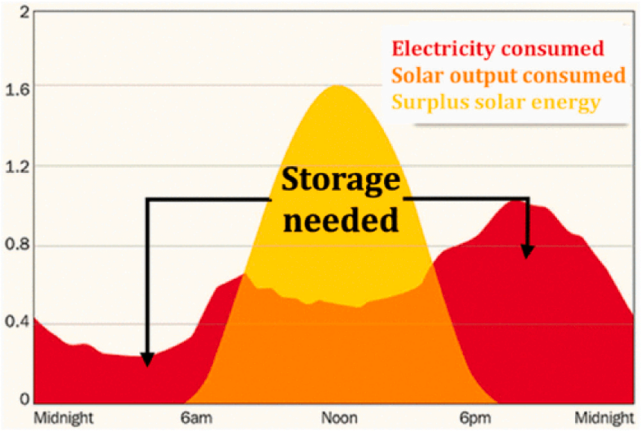


Fig. 20. A graphic showing the potential for solar energy generation to reduce peak usage and the common need for energy storage solutions. Image acquired from Carrillo et al (Carrillo et al., 2019).

While electrochemical batteries are key in many systems, scheduling energy use can have similar effects without needing high capital investments by matching the electricity demand to the supply. The flexible scheduling of operations studied by Weitzel and Glock showed a peak demand reduction of 60 %, only a 20 % decrease from a similar situation with BESS (Weitzel and Glock, 2019). The payback period of a pumped water system decreased from 17 to 10.4 when a battery was replaced by a process of pumping water when excess solar was produced (Chen et al., 2024). Lastly, Ruiz, Fan, and Jin achieved a 10 % monthly cost savings using only PV and scheduled production; adding a BESS increased the savings by only 12 % (Ruiz Duarte et al., 2020).

Table 15 shows that solar PV electricity generation can be one of the most scalable generation techniques, with arrays ranging from 0.3 to 6000 kW in size. The peak demand reductions follow suit, ranging from 0.2 % to 60 or even 80 %. However, the size and percentage of peak reduction are often inversely related. Fig. 21 depicts the percentage of demand shifted and the solar array size for the compiled papers, where available. It shows that large demand shifts are available only when the facility and array are small. For bigger facilities, PV or PV and energy storage can only achieve low demand shifting compared to smaller facilities.

6.2.2. Wind

Wind onsite energy generation can effectively offset electricity

Table 15
Peak Reductions, Saving, and Synergistic Technologies of Photovoltaic Systems from Select Studies.

Papers used	(Weitzel and Glock, 2019; Fleschutz et al., 2022; Nan et al., 2021; Kusakana, 2020; Yun et al., 2022; Ruiz Duarte et al., 2020; Saffari et al., 2018; Arteconi et al., 2017; Chen et al., 2024)	
	Median	**IQR**
Demand Shifted or Reduced (kW)	100	85 – 331
Percent of Demand Shifted or Reduced (%)	11	5.1 – 49
Energy Shifted or Reduced (kWh)	700	125 – 10,000
Percent of Energy Shifted or Reduced (%)	3.5	3.5 – 13
Payback Period (years)	16.5	14.35 – 25.55
Monetary Savings (USD per year)	19,000	9,000 – 269,000
Percent of Monetary Savings (%)	5	4.1 – 10
Size (kW)	350	20 – 6000
Price (USD)	7,627,000	4,233,000 – 7,927,000

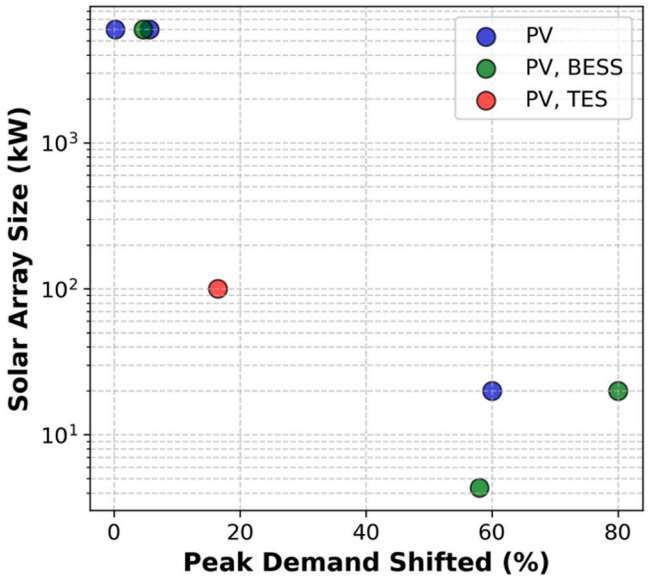


Fig. 21. Solar array size vs. the percentage of the peak demand shifted for the compiled papers. Original image.

demand and enhance grid efficiency in industrial settings. However, because of the nature of effective wind farms, this typically has to be done at a larger scale, incorporating a microgrid with multiple industrial facilities. Because the existing demand from multiple facilities is so large, storage devices are less critical than they are with PV. Wind has separate generation patterns from solar. While solar provides increased capacity during sunny days, wind can help electricity loads overnight and through winter months due to the daily and seasonal trends, as shown in Fig. 22 and Fig. 23 (Edrisian et al., 2013; Zakariazadeh et al., 2014).

Wind generation typically occurs on the MW scale, reducing peak electricity demand by 40 MW at times, as demonstrated by Falsafi (Falsafi et al., 2014). By leveraging price-based and incentive-based DR programs, another facility was able to account for a 6 MW reduction in power demand due to wind generation, equating to 13 % of the total load (Xu et al., 2019). Because of the volatility of wind power, integration can also be supplemented by operational monitoring and flexibility. Using a day-ahead pricing scheme, a demand reduction tailored to the aluminum, steel, and cement industries showed a 20 % decrease on the magnitude of 4 MW (Gerami et al., 2021). Additionally, a stochastic scheduling model integrated wind and natural gas generation to

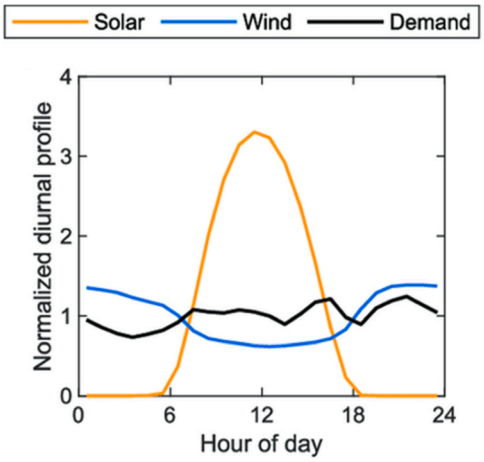


Fig. 22. Typical power generation vs consumption for solar and wind outputs. Image acquired from Song et al (Song et al., 2022).

optimize resource dispatch and manage renewable variability. This hybrid solution achieved a 68 % reduction in power demand with a cost savings of 3.4 % in a 25 MW system (Zhang et al., 2016c). Table 16 summarizes these findings for wind energy generation, showing that the peak demand shift from these results ranges from 13 % to 68 % and 4–40 MW.

6.2.3. *Hybrid*

Combining PV and wind energy, hybrid onsite energy generation offers a robust solution to offset electricity demand in many residential and commercial sites (Patel and Beik, 2021; Dahiru et al., 2023). However, research on its application in the industrial sector remains limited. The advantage of hybrid energy generation is that it allows the integration of AC and DC currents, as shown in Fig. 24. This integration, coupled with advanced energy storage and/or other IDR strategies, enhances the reliability and sustainability of industrial operations. By using a 230 kW solar array and a 300 kW wind farm, Wang et al. demonstrated a decrease in energy usage of 945 kWh/day (though the specifics of peak shavings were not shared) (Wang et al., 2017). Hybrid generation can also be combined effectively with other generation methods. To address challenges in renewable energy variability, a novel "value storage" concept was proposed by Almehizia et al. This approach converts excess renewable energy into storable industrial products (e.g., steel or aluminum), eliminating reliance on BESS. The hybrid system combined PV generation, wind energy, pumped water storage, and thermal storage, achieving over 50 % demand reduction (Almehizia et al., 2019). The peak demand for a facility can even be entirely supplied, as shown in a case study involving a lithium battery manufacturing plant. Pei et al. integrated hybrid generation, compressed air, and thermal energy storage to optimize energy usage, achieving a 163 % reduction in peak energy usage from the grid (selling excess back to the power provider). Different implementation strategies achieved cost savings of between 2.6 % and 26 % (Pei et al., 2019).

Because of the large size of these projects, savings were significant across all studies, as shown in Table 17. Though the size makes this hybrid energy generation much harder to implement at an industrial scale, the biggest strengths include decreased costs through energy savings and high renewable energy penetration in the industrial sector.

6.3. *Summary of Energy Generation Solutions*

Energy generation technologies can significantly affect both the energy and demand of a facility's grid reliance. While this paper focused on cost reduction due to reduced demand, both demand and energy savings help increase the financial success of these technologies. This can be coupled with environmental impacts, as fewer scope two emissions are released due to increased efficiency or the complete displacement by renewable energies. Fig. 25 and Fig. 26 outline the percentage of load shifted, the payback period, and the impact of timing on the facility's energy use for the various storage solutions. The impact on energy use spans a wider range than other solutions presented in this paper. Depending on the size, type, and involvement of energy storage, some cases require no change to the facility's timing of energy use. In contrast, others require moderate changes to production times/procedures, especially those trying to use energy arbitrage. Considering load reduction, energy generation is the only way to reduce 100 + % of the peak demand. Similar to storage systems, these technologies typically have longer payback periods. Unlike storage systems, the payback can be shorter due to multiple significant revenue streams. Electricity generation typically results in low-to-medium impact but medium-to-high payback periods.

7. Intelligent Operations

Many solutions presented in this paper increase savings with intelligent operations. These include predictive analytics, forecasting, real-

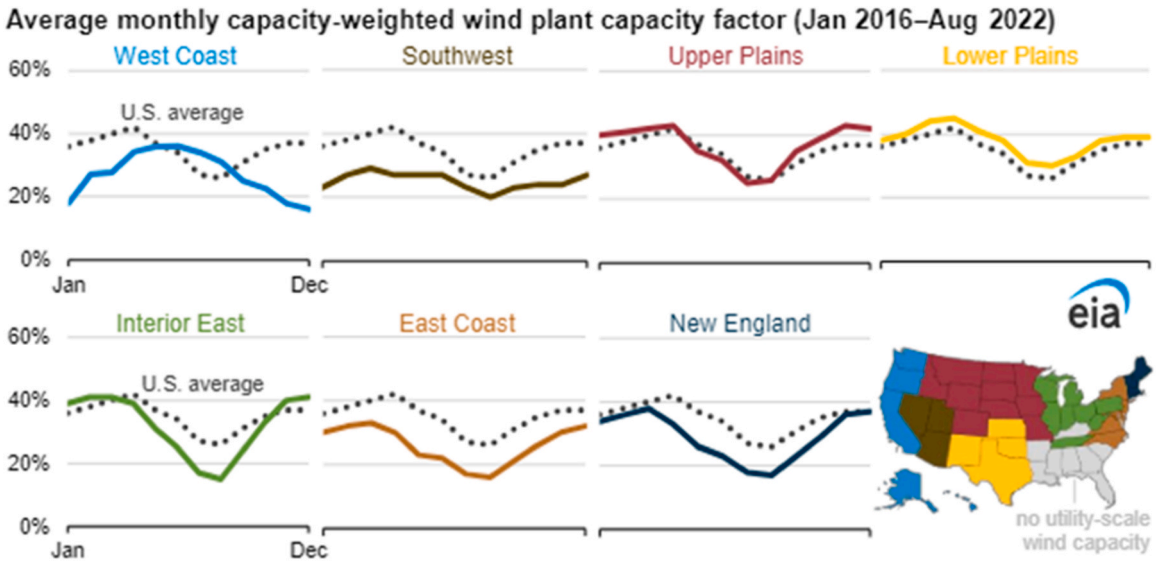


Fig. 23. General seasonal trends for solar and wind generation. Image acquired from Brown at EIA (Brown, 2022).

Table 16
A Summary of Wind Systems from Select Studies.

Paper Title	Demand Shifting Method	kW Shifted /Reduced	Savings Mechanism	Monetary Savings (per year)	Size (kW)
(Falsafi et al., 2014)	Wind Generation	6,000 13 %	Energy costs	$306,600 19 %	50,000
(Xu et al., 2019)	Wind Generation	40,000	Energy costs		160,000
(Gerami et al., 2021)	Wind Generation, Real-Time Scheduling, PAF	4,000 20 %	DRP	$468,000	19,000
(Zhang et al., 2016c)	Wind Generation, Gas Generation, Real-Time Monitoring	17,000 68 %	Energy costs, demand costs	$669,000 3.4 %	115,000

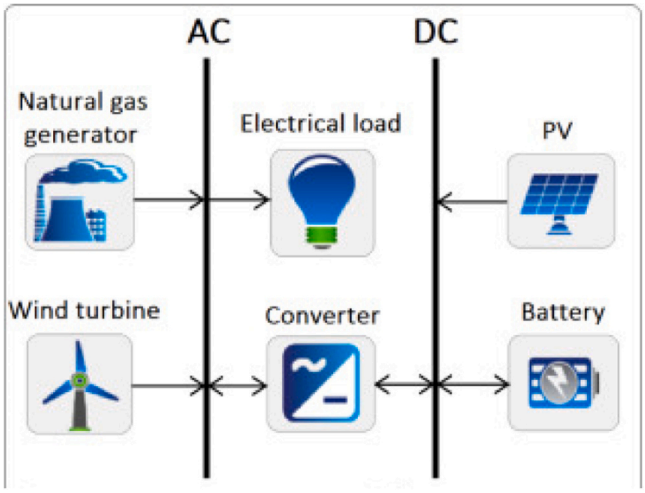


Fig. 24. An example of how solar (DC generation), wind (AC generation), storage, and electrical loads work together. Image acquired from Khosravani et al (Khosravani et al., 2023).

time monitoring, and optimization. While no direct savings result from these mechanisms, they enable increased savings when combined with the other technologies presented thus far. Likewise, all citations used in this section have already been mentioned in the above content, but this section highlights the increased performance when combined with intelligent operations. While the savings percentages range greatly depending on the application, this section outlines the types of systems that can benefit from intelligent operations, showcasing their widespread use and advantages in different situations.

7.1. *Predictive Analytics and Forecasting*

Predictive analytics and forecasting (PAF) have emerged as essential tools for optimizing IDR. These technologies use facility data and models to predict energy use, energy generation, and dynamic pricing. If future demand profiles can be predicted, changes in the current process can be used to minimize peak demand or maximize cost savings (Wang et al., 2021a, 2021b; Walther et al., 2019). An example of an electric forecaster and prediction is shown in Fig. 27, illustrating the accuracy obtainable by some models.

Predictive analytics is key in optimizing energy storage, such as BESS and TES. A depleted storage system cannot reduce peaks, but PAF can forecast energy and demand needs, enabling optimal charging strategies. For example, a cement crushing plant used MPC with BESS to shift up to 2 MW of demand, reducing peak energy use by 20 % (Zhang et al., 2016a). Optimization also aided the decrease in payback in a food processing plant, paying for a BESS in only 7–9 years (Wohlgenannt et al., 2024). Machalek and Powell compared active and passive TES in an industrial bakery, highlighting a 4–12 % reduction in peak demand and a 1–3.3 % cost savings through real-time adjustments (Machalek and Powell, 2019a).

Another innovative application is accompanied by preplanned or real-time scheduling, allowing precise alignment of production schedules with energy market fluctuations. Tsay et al. employed low-order Hammerstein-Wiener models to optimize the scheduling of an air separation unit, achieving over 3–4 % demand savings and 13 % cost savings in real-time energy markets (Tsay et al., 2019b). Similarly, Lu et al.

Table 17
A Summary of Hybrid Generation Systems from Select Studies.

Paper Title	Demand Shifting Method	kW Shifted /Reduced	kWh Shifted /Reduced	Savings Mechanism	Monetary Savings (per year)
(Wang et al., 2017)	Hybrid Generation, Other Storage		945 / day	Energy costs, demand costs	$31,000
(Almehizia et al., 2019)	Hybrid Generation, Pumped Hydro Storage, Thermal Energy Storage	50 %		Energy costs	$63 M 10 %
(Pei et al., 2019)	Hybrid Generation, Compressed Air Storage, Thermal Energy Storage, CHP, Predictive Analytics	6,000, 6,500 150, 163 %		Energy costs	2.60, 25.5 %

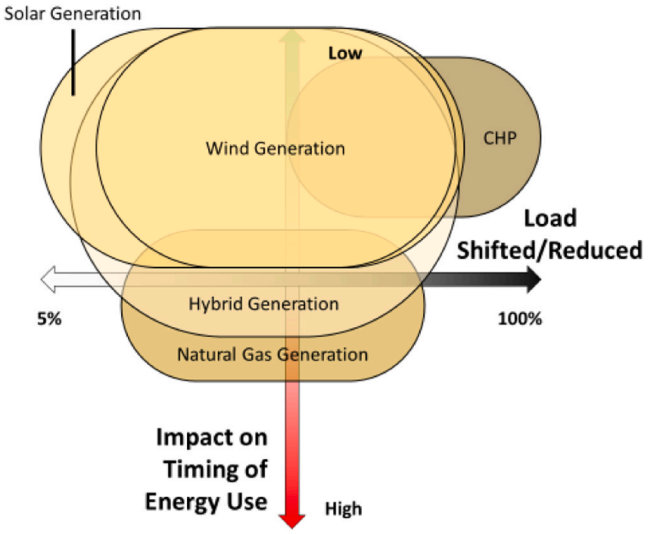


Fig. 25. Impact on timing of energy use vs. load shifted or reduced for energy generation. Original image.

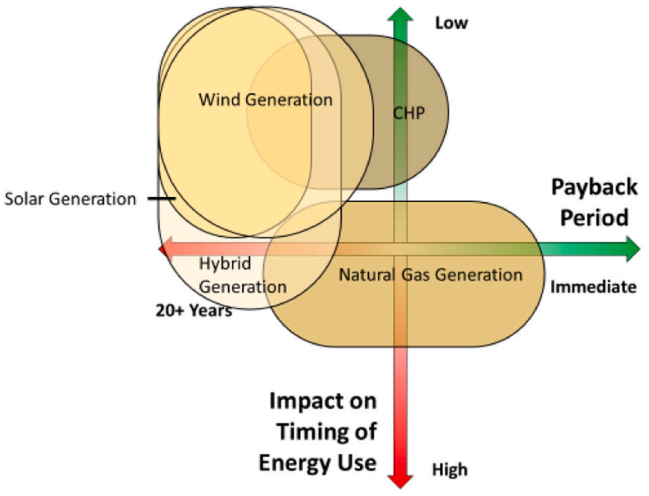


Fig. 26. Impact on timing of energy use vs. payback period for energy generation. Original image.

utilized a long short-term memory neural network for real-time price--based IDR, enabling load shifts of up to 50 kW during off-peak periods and a 13.31 % reduction in electricity costs (Lu et al., 2021). DEC can assist in saving when machinery cannot be shifted away from peak hours. A rotary kiln used for limestone calcining adjusted critical variables such as rotation and feed rates, allowing the kiln to reduce demand by 30–70 % during grid fluctuations (Machalek and Powell, 2019b). Similar numbers were seen in mine operations also controlled by DEC. Moving pumping to off-peak times using PAF allowed a 76 % decrease in peak energy usage (Machalek et al., 2020a).

Finally, PAF enhances the integration of energy generation into industrial processes. This can either be through the prediction of process needs or pricing predictions, allowing generation to ramp up when needed. Though not completely accurate, pricing models can induce savings despite errors. In aluminum, steel, and cement industries, the overall load was decreased by 17.75 %, and cost savings were increased by 41.3 % by modeling renewable generation from wind (Gerami et al., 2021). Depending on the size, control scheme, and daily pricing ranges, savings between 2.6 % and 31 % have been reported on CHP and natural gas systems (Pei et al., 2019; Cui et al., 2016).

Table 18 shows the variety of applications that can be improved by using PAF, integrating itself with all other sections presented in this paper. The range of savings due to PAF is typically between 5 % and 15 % but requires a high need for flexibility in operation. Without flexibility, there is no reason to forecast the future. With the only costs coming from sensors or time to operate and maintain PAF, this technology finds itself in IDR solutions' low-cost, high-flexibility area.

7.2. *Real-Time Monitoring*

Real-time monitoring (RTM) technologies enable industrial facilities to observe and respond to energy use patterns as they occur, improving the effectiveness of IDR. These technologies require and use real-time data, whether on how the system uses electricity or real-time pricing in the market space. This data is then integrated with energy storage systems, dynamic controls, and smart grid environments to optimize energy usage, reduce peak demand, and align operations with fluctuating electricity prices.

The most common use of RTM is to monitor the facility's energy usage and to control storage devices dynamically. One approach involves BESS to reduce peak demand by up to 20 % (2 MW) at a cement crushing plant using TOU pricing mechanisms (Zhang et al., 2016a). Similarly, wastewater treatment plants are excellent candidates for real-time IDR programs because of their flexible load capabilities. Using CHP generation, battery storage, and dynamic flow rate controls, Musabandesu and Loge demonstrated how these plants could shift up to 15 % of peak demand and 1 MWh of energy consumption (this increased to 30 % of peak load when combined with CHP) (Musabandesu and Loge, 2021). In addition to BESS, TES offers viable solutions for real-time load management. A facility integrating TES with VFD chillers and RTM achieved a 5–10 % peak reduction and a 6.5-year payback (Bahr et al., 2023). Pumped water storage allows for easy integration, allowing pumps to fill tanks during off-peak hours. This can result in a 4–8 % peak reduction with little additional capital cost for sites utilizing process water (Fekri Moghadam et al., 2015; Chen et al., 2024). RTM can also help large systems, which aided a combination of solar, wind, and natural gas generation to lower peaks by 68 % (17 MW) (Zhang et al., 2016c).

RTM also allows for real-time response in scheduling practices. For instance, Westberg et al. implemented proactive automation in a batch manufacturing facility, staggering mixer schedules and utilizing glycol storage tanks as temporary energy reserves. Various scheduling methods resulted in a peak reduction of 11–19 % or a cost savings of 5.3–14 %, depending on the scheduling method (Westberg et al., 2019). A slightly

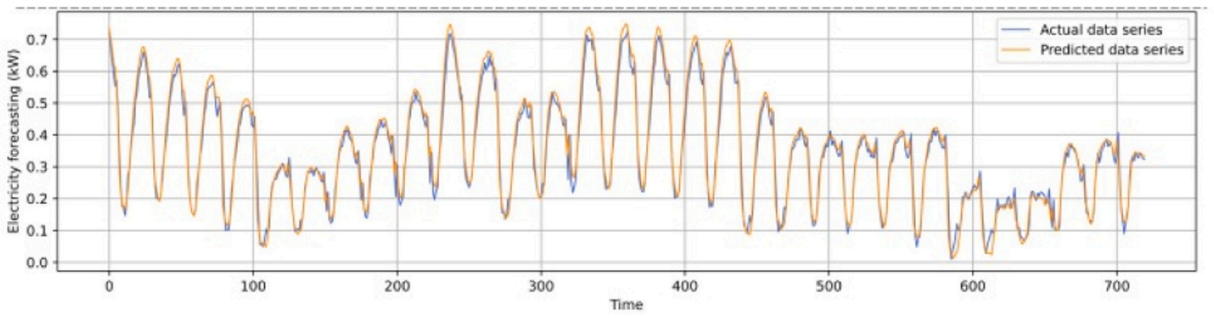


Fig. 27. An illustration of energy usage predicted through modeling vs. the actual usage later recorded by the facility. Image acquired from Khan et al (Khan et al., 2022).

Table 18
Peak Reductions, Saving, and Synergistic Technologies of Predictive Analytics and Forecasting from Select Studies.

Papers used	(Zhang et al., 2016a; Wohlgenannt et al., 2024; Machalek and Powell, 2019a; Pei et al., 2019; Lu et al., 2021; Machalek et al., 2020a; Simkoff and Baldea, 2020; Gerami et al., 2021; Machalek and Powell, 2019b; Cui et al., 2016; Tsay et al., 2019b)	
	Median	**IQR**
Demand Shifted or Reduced (kW)	780	37.5 – 6,310
Percent of Demand Shifted or Reduced (%)	20.5	10.5 – 68.3
Energy Shifted or Reduced (kWh)	60.7	39.3 – 168
Percent of Energy Shifted or Reduced (%)	30	19.3 – 44.3
Monetary Savings (USD per year)	8,600	425 – 13,000
Percent of Monetary Savings (%)	9	2.6 – 13.3

more significant reduction of 29 % in the peak energy usage occurred in a chlor-alkali plant implementing real-time scheduling in response to an RTM method (Otashu and Baldea, 2019). This response can help when attempting to respond to DRP. An HVAC system in a manufacturing facility, using real-time data to reduce energy usage temporarily during DR events, saw energy cost savings of $100–$300 per event (Brundage et al., 2014).

The results of RTM, as shown in Table 19, are similar to that of PAF. RTM is another method that integrates itself nicely with other solutions, enabling an extra 5–10 % in peak reduction or cost savings. The costs of this technology are also similar: primarily costs surrounding sensors, costs to develop algorithms, or operational costs.

Table 19
Peak Reductions, Saving, and Synergistic Technologies of Real-Time Monitoring from Select Studies.

Papers used	(Zhang et al., 2016a; Musabandesu and Loge, 2021; Bahr et al., 2023; Fekri Moghadam et al., 2015; Chen et al., 2024; Westberg et al., 2019; Brundage et al., 2014; Otashu and Baldea, 2019; Zhang et al., 2016c)	
	Median	IQR
Demand Shifted or Reduced (kW)	250	80 – 1,750
Percent of Demand Shifted or Reduced (%)	11	7.8 – 20
Energy Shifted or Reduced (kWh)	750	200 – 109,000
Percent of Energy Shifted or Reduced (%)	3.9	2.2 – 4.88
Monetary Savings (USD per year)	19,500	7,750 – 127,000
Percent of Monetary Savings (%)	5.3	4.1 – 6.3

7.3. Optimization

Optimization is another powerful intelligence mechanism in DR strategies. Some solutions mentioned in this paper were put into place without any advanced assessment but rather 'eyeballed' by the researcher or facility. By leveraging advanced algorithms, industrial facilities can further reduce their energy consumption while minimizing operational costs (Röder et al., 2024).

Optimization can help use energy storage solutions to their full extent. Using optimization to utilize TES has proven useful for smaller TES applications – 30–150 kW savings (Machalek and Powell, 2019a; Bahr et al., 2023) – and larger systems – 3–4 MW savings (Molavi and Ardehali, 2016). Storage in ammonia plants specifically has used optimization and production schedules, resulting in demand reduction of up to 7.4 % and cost savings between 1.39 % and 3.7 % (Kelley et al., 2022a).

A significant use of optimization is to schedule production, cycling, startup, and even curtailment, which is also very scalable. Small facilities such as sugar factories and air separation units have used optimization in situations with peaks around 250 kW (Rui et al., 2024; Pablos et al., 2021; Tsay et al., 2019a). In more significant demand situations, such as those found in refinery processes, cement plants, and other large users, optimization has helped diminish peaks on a magnitude of 5–15 MW (Kholerdi and Ghasemi-Marzbali, 2021; Reka and Ramesh, 2016; Zhao et al., 2014). Using optimization to cycle production in a facility can also have tremendous savings, often decreasing peak load by 70–80 % during demand response events or times of high electricity price (Lu et al. specifically used a Multi-agent deep deterministic policy gradient algorithm) (Desta et al., 2018; Lu et al., 2020). Combining scheduled and cycled production, a deep reinforcement learning strategy saved a steel powder manufacturer 24 % of their daily electricity cost (Huang et al., 2019). Other optimization techniques, including genetic algorithms and mixed-integer linear programming, were applied to scheduled startup scenarios, resulting in demand and cost savings of around 20 % (Brimley et al., 2019; Gong et al., 2015). While the savings only amounted to 1–2.7 %, optimization has also been used in curtailment situations (Pattison et al., 2016; Kelley et al., 2018; Baldea, 2020).

Another successful strategy is to use real-time optimization, allowing a facility to participate in real-time electricity pricing. When switching from pre-planned to real-time scheduling in a flour mill, peak demand saw a further demand reduction from 13 % to 70 % and increased cost savings by 7 % (Zaidi et al., 2020). Similarly, for air separation units, the cost savings increased from 3 % to 13 % for a switch from preplanned to real-time curtailment, though the peak reduction only increased slightly (Tsay et al., 2019a).

Lastly, optimization frameworks extend to dynamic control of natural gas energy generation, allowing flexibility in energy source usage. While the main benefits of this kind of electricity generation come from the nature of the solution, optimization can often add 5–10 % savings from the base case. This reduction proves true for a variety of system

Table 20
Peak Reductions, Saving, and Synergistic Technologies of Optimization from Select Studies.

Papers used	(Molavi and Ardehali, 2016; Machalek and Powell, 2019a; Bahr et al., 2023; Kelley et al., 2022a; Rui et al., 2024; Pablos et al., 2021; Kholerdi and Ghasemi-Marzbali, 2021; Zaidi et al., 2020; Reka and Ramesh, 2016; Lu et al., 2021; Zhao et al., 2014; Huang et al., 2019; Desta et al., 2018; Lu et al., 2020; Brimley et al., 2019; Gong et al., 2015; Henning et al., 2019; Tsay et al., 2019a; Pattison et al., 2016; Kelley et al., 2018; Baldea, 2020; Seo et al., 2020; Dababneh and Li, 2019; Cui et al., 2016)	
	Median	**IQR**
Demand Shifted or Reduced (kW)	120	37.5 – 5,220
Percent of Demand Shifted or Reduced (%)	11	8.4 – 20
Energy Shifted or Reduced (kWh)	280	150 – 560
Percent of Energy Shifted or Reduced (%)	25.9	2.9 – 55.6
Monetary Savings (USD per year)	15,700	3,760 – 36,900
Percent of Monetary Savings (%)	13	3.96 – 31

sizes, from 600 kW to tens of MW (Seo et al., 2020; Dababneh and Li, 2019; Cui et al., 2016)

Table 20 shows a similar summary for optimization as compared to PAF and RTM. However, because optimization techniques can be used in the design phase of implementing other solutions and during the real-time use of electricity, the attributed peak reduction percentage increases slightly, ranging from 5 % to 20 %. By employing diverse optimization strategies, industrial facilities can effectively navigate energy market challenges, minimize operational costs, and contribute to grid stability.

7.4. Summary of Intelligent Operations

When the technologies presented in this paper are combined with intelligent operations, they achieve higher demand, energy, and/or cost savings than their non-intelligent counterparts. The savings percentages are typically between 5 % and 10 %. Additionally, this enabled savings opportunities that would not have been available, especially when implementing real-time monitoring. Fig. 28 and Fig. 29 outline the percentage of load shifted, the payback period, and the impact of timing on the facility's energy use for the various storage solutions. These statistics are more challenging to compile as there must be a base case and a case using intelligent operations to calculate the difference. The load shifted and payback periods for these figures are based on quantitative numbers from references, while the impact is set with a qualitative description of each storage solution. This shows the typical lower impact but higher payback periods of these technologies.

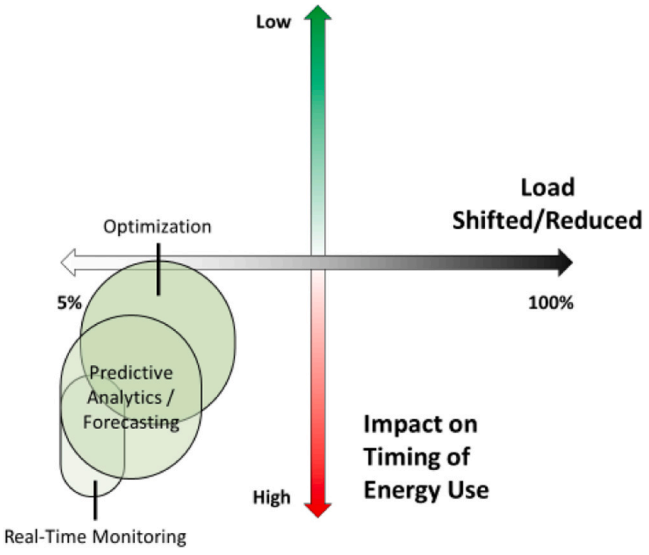


Fig. 28. Impact on timing of energy use vs. load shifted or reduced for intelligent operations. Original image.

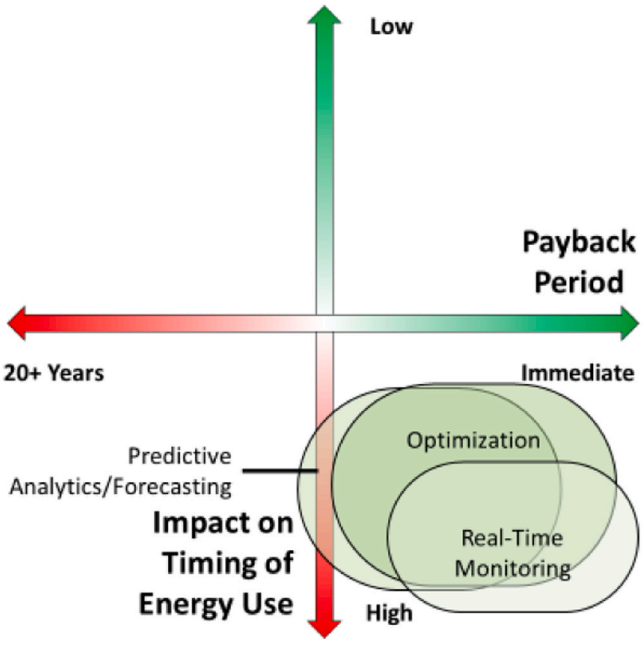


Fig. 29. Impact on timing of energy use vs. payback period for intelligent operations. Original image.

8. Conclusion

Energy storage allows little to no impact on energy use timing. In some situations with larger financial incentives, storage solutions offer cost benefits within the lifetime of the technology. However, this is not always the case, sometimes resulting in paybacks of 20 + years and rendering the technology unreasonable without added benefits of protection from blackouts, process resiliency, or process electrification. On-site energy generation is another low-impact, high-payback area. Unlike energy storage, generation solutions typically offer larger financial incentives, lowering the overall utility price for the facility. Energy generation requires coordination with the facility's energy usage, resulting in slightly more impact on the timing of energy use.

In the other area, high impact but low payback, rests the technologies of scheduled energy usage, operational flexibility, and intelligent operations. Scheduled energy use is a very effective method for reducing peak demand and can be implemented with little to no additional costs. However, it requires moderate flexibility and is not an option for all facilities, namely those with non-schedulable tasks. A similar thing can be said for operational flexibility. Because of the greater need for flexibility, it may be even harder to implement for facilities. However, these solutions allow manufacturers to take advantage of real-time decision making or pricing markets. Intelligent operations do not create savings themselves, but they can add a 5–20 % peak reduction or savings to existing projects. They supplement various projects very well, but require variable parameters that can be changed to create an optimization problem.

IDR technologies typically camp in these two areas - solutions with high payback periods but lower impact on the timing of an industrial

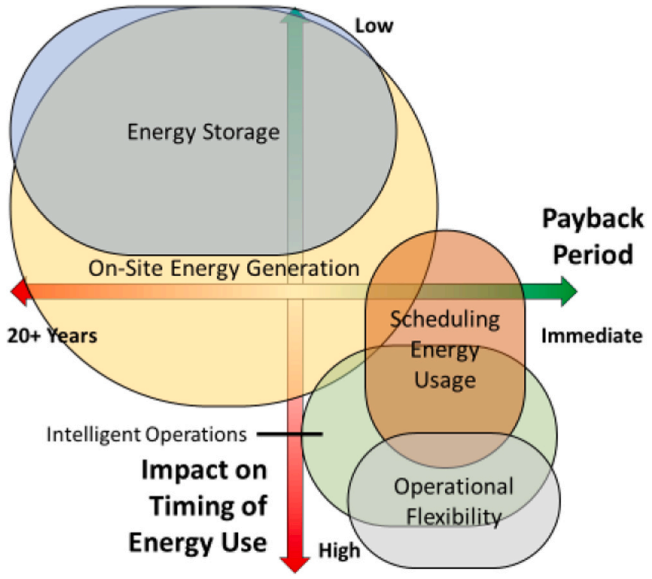


Fig. 30. Overview of impact on timing of energy use vs. payback period for all technologies. Original image.

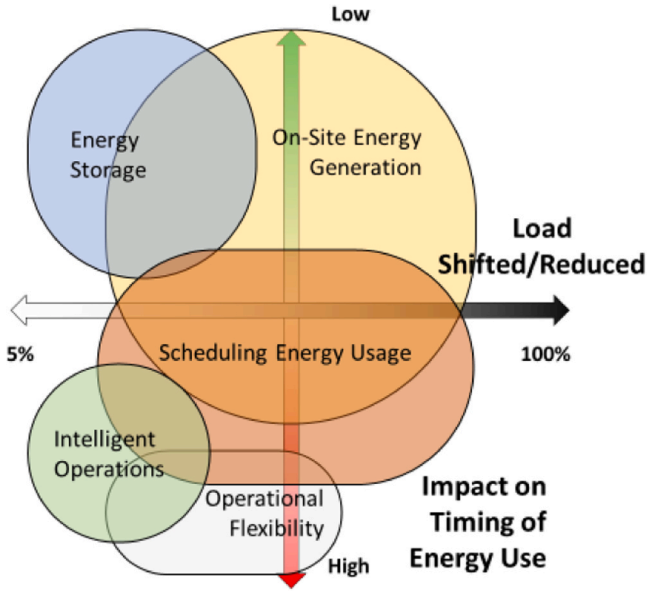


Fig. 31. Overview of impact on timing of energy use vs. load shifted or reduced for all technologies. Original image.

facility's energy use, or solutions with low payback but high impact on energy use - as depicted in Fig. 30. The two areas of IDR that approach the top right corner (the optimal location) are on-site energy generation and scheduled energy usage. One reason this may be the case is the ability to shift large percentages of the peak load, as seen in Fig. 31. The larger the percentage of the shifted load, the larger the available savings. Another reason may be that these technologies collaborate better with others, allowing the advantages of separate technologies to combine for greater success.

It is also important to see the interaction of certain technologies together. Fig. 32 depicts the number of studies that use different types of solutions. For example, 28 studies used multiple storage solutions, and 25 used both storage and energy generation (such as electrochemical batteries and PV cells). The figure highlights the frequent need for multiple solutions and reveals which combinations are the most practical and widely adopted. The most notable connections are the two

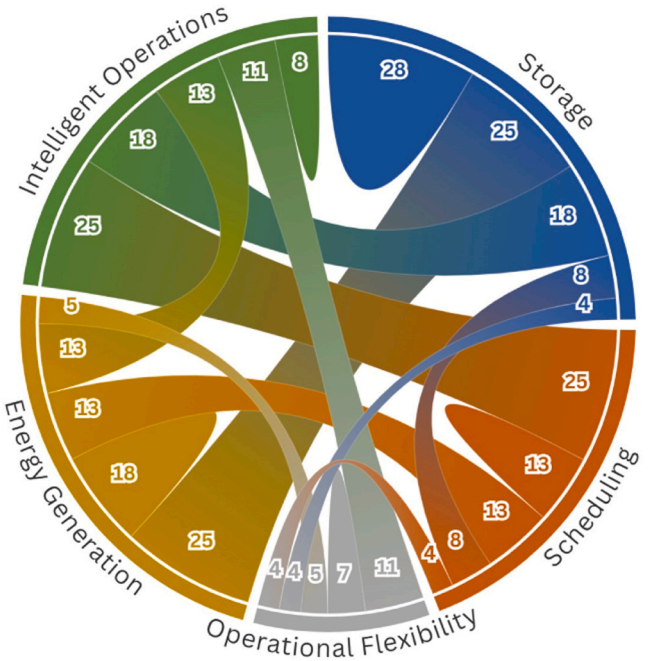


Fig. 32. A chord diagram depicting the number of technology interactions, where each chord is the number of studies using both solutions to achieve peak reduction. Original image.

aforementioned combinations, as well as scheduled energy use being joined with intelligent operations.

The need for electricity is increasing in the industrial sector. This increase is creating problems such as grid instability and rising costs. Additionally, advancements in electricity generation are heading towards more renewable resources, leading towards the electrification of energy generation and away from on-site natural gas use. While the industrial sector has lagged in the past, IDR is emerging as a strong solution to address these issues in the industrial sector. While all of these technologies and solutions are not applicable to every situation, they portray the plethora of options and innovations that can be applied to industrial processes in one way or another.

Many hurdles remain to achieve large-scale implementation of IDR solutions, especially those capable of large percentages of reduction in the industrial sector. However, these examples of demand shifting show the economic and environmental advantages already present in industrial facilities. Many of these successes are due to the unique energy usage of manufacturing facilities, making unique solutions that are not applicable or successful in residential or commercial sectors. As the grid continues to search for solutions that will enhance grid stability, integrate renewable energies, electrify energy generation, and lower overall costs, additional effort spent creating a flexible demand in the industrial sector can continue to aid all of these goals.

8.1. *Limitations and Direction for Future Research*

This review is limited by the heterogeneity of the underlying literature. Reported payback periods and peak reductions are drawn from studies that vary widely in tariff structures, industrial sectors, currencies, and methodological approaches. As a result, the ranges synthesized here should be interpreted as indicative rather than universally comparable. In addition, the evidence base is uneven across regions and industries, with a concentration of case studies from North America and heavy manufacturing, leaving gaps in knowledge for other geographies and sectors. Because of the large impact a singular, large industry may have on the grid, smaller facilities are often excluded from IDR studies and therefore have less of a representation in this review.

As highlighted in this review, many economically viable demand response solutions depend on the degree of operational flexibility available within industrial processes. Likewise, the structure of electricity tariffs strongly influences the effectiveness of these strategies. Further research is needed to examine how different pricing mechanisms shape the feasibility and performance of IDR, thereby clarifying both the range of tools facilities can deploy and the economic conditions under which they are most effective. Because geographical location has a smaller impact on load structure and greater impact on pricing structure, insights into pricing structures would also help investigation into IDR be better transferred to different geographical locations. In addition, synergistic strategies represent a promising area for future study. By combining complementary approaches, it may be possible to reduce the flexibility required from individual processes while achieving superior economic outcomes, creating pathways to more robust and scalable implementations of industrial demand response.

CRediT authorship contribution statement

Matthew DeHaan: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Powell Kody:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization. **Julie Sieving:** Writing – review & editing, Visualization, Supervision. **Blake Billings:** Writing – review & editing, Supervision, Resources, Data curation.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT in order to improve the readability and language of the manuscript. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.egyr.2025.108962.

Data availability

Data is available in the related papers cited in the review.

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